

1 Introduction

This document contains textbook questions to warm up as well as HSC questions from the past 5 years categorised in each category. I want you guys to use this document to guide your consolidation of past concepts. You should aim to finish this question bank, it seems like a lot but if you plan your routine well, i.e. completing parts of it at regular intervals then you will get through it quickly no doubt. Eventually, I want you guys to take the exams in full test conditions and have you guys confidently answer every single question. This question bank should hopefully provide the first step. I haven't written any notes or given any hints however I can provide a complementary notes document if you guys think it will be a good idea. I just hope you guys have taken good notes and now see the value in quality note taking. Good luck, try to attempt every question, however if you find you understand a concept to a well enough degree, you may skip those questions for the sake of boredom. You can click on the questions using the table of contents below, hopefully it will make your navigation of this document easier.

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2 Kinematics and Dynamics

2.1 Equations of Motion

2.1.1 Question 1 (Textbook)

Distinguish between a vector and a scalar.

2.1.2 Question 2 (Textbook)

What is the different between an average quantity and an instantaneous quantity? Can you give an example?

2.1.3 Question 3 (Textbook)

An airplane lands with an initial velocity of 70.0 m/s and then decelerates at 1.50 m/s² for 40.0 s. What is its final velocity?

2.1.4 Question 4 (Textbook)

A manned rocket accelerates at a rate of 20 m/s² during launch. How long does it take the rocket reach a velocity of 400 m/s?

2.1.5 Question 5 (Textbook)

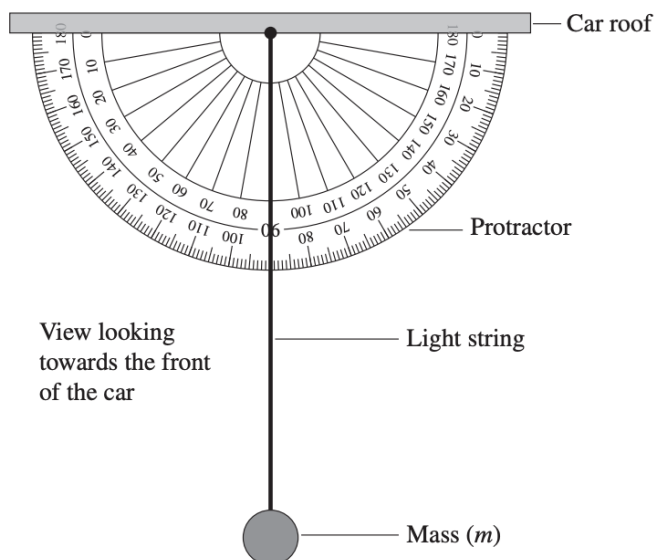
A soccer player starts from rest and accelerates forward, reaching a velocity of 8.00 m/s in 2.50 s. (a) What was his average acceleration? (b) What average force did he exert backward on the ground to achieve this acceleration? The players mass is 70.0 kg, and air resistance is negligible.

2.1.6 Question 6 (HSC 2019 Q35)

Question 35 (4 marks)

The apparatus shown is attached horizontally to the roof inside a stationary car. The plane of the protractor is perpendicular to the sides of the car.

4



The car was then driven at a constant speed (v), on a horizontal surface, causing the string to swing to the right and remain at a constant angle (θ) measured with respect to the vertical.

Describe how the apparatus can be used to determine features of the car's motion. In your answer, derive an expression that relates a feature of the car's motion to the angle θ .

2.2 Vectors and Forces

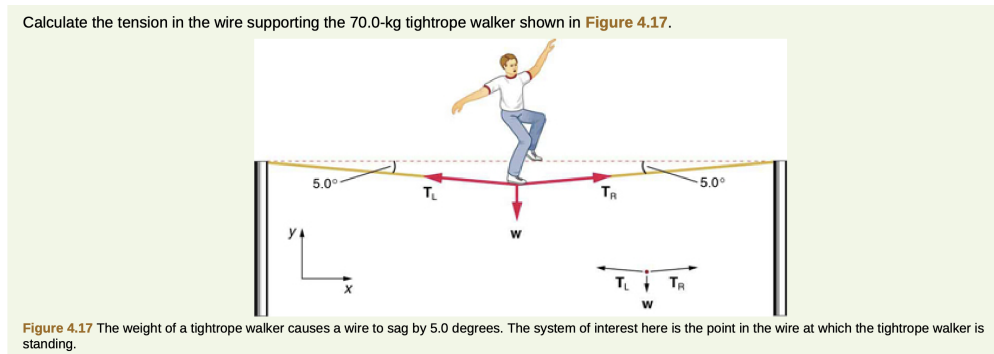
2.2.1 Question 1 (Textbook)

Use the graphical technique for adding vectors to find the total displacement of a person who walks the following three paths (displacements) on a flat field. First, she walks 25.0 m in a direction 49.0° north of east. Then, she walks 23.0 m heading 15.0° north of east. Finally, she turns and walks 32.0 m in a direction 68.0° south of east.

2.2.2 Question 2 (Textbook)

Find $\vec{A} + \vec{B}$. Vector A represents the first leg of a walk in which a person walks 53.0 m in a direction 20.0° north of east. Vector B represents the second leg, a displacement of 34.0 m in a direction 63.0° north of east.

2.2.3 Question 3 (Textbook)

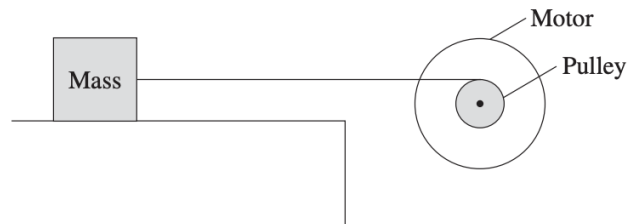


2.2.4 Question 4 (HSC 2020 Q32)

Question 32 (7 marks)

A rope connects a mass on a horizontal surface to a pulley attached to an electric motor as shown.

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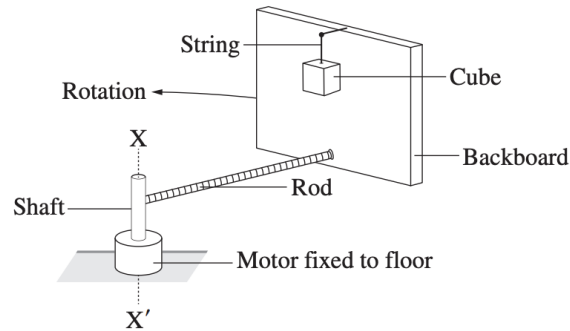


Explain the factors that limit the speed at which the mass can be pulled along the horizontal surface. Use mathematical models to support your answer.

2.2.5 Question 5 (HSC 2019 Q20)

- 20 In the apparatus shown, a backboard is connected by a rod to a shaft. The shaft is spun by an electric motor causing the backboard to rotate in the horizontal plane around the axis $X-X'$.

A cube is suspended by a string so that it touches the surface of the backboard.



When the angular velocity of the motor is great enough, the string is cut and the position of the cube does not change relative to the backboard.

Which statement correctly describes the forces after the string is cut?

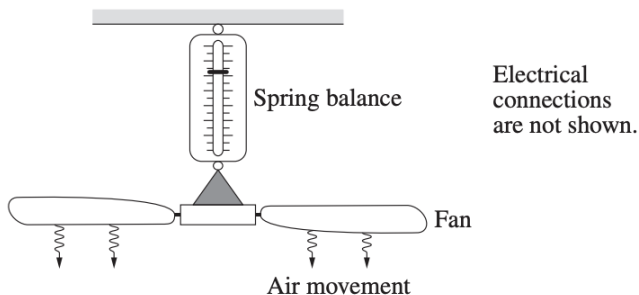
- A. The sum of the forces on the cube is zero.
- B. The horizontal force of the backboard on the cube is equal in magnitude to the horizontal force of the cube on the backboard.
- C. The horizontal force of the backboard on the cube is greater than the horizontal force of the cube on the backboard, resulting in a net centripetal force.
- D. The force of friction between the cube and the backboard is independent of the force of the backboard on the cube because these forces are perpendicular to each other.

2.2.6 Question 6 (HSC 2019 Q31)

Question 31 (8 marks)

A student suspends an electric ceiling fan from a spring balance.

The fan is switched on, reaching a maximum rotational velocity after ten seconds.



- (a) Explain the changes that would be observed on the spring balance in the first 15 seconds after the fan is switched on. 4

2.3 Work and Energy

2.3.1 Question 1 (Textbook)

Distinguish between work and energy. Why do we have a separate term known as work even though the units are the same?

2.3.2 Question 2 (Textbook)

What is the conservation of energy and how does it relate to the work done on a system with total energy E , can you give an example using equations?

2.3.3 Question 3 (Textbook)

When should we use $U = mgh$ vs $U = -\frac{GMm}{r}$, why is U considered to be negative?

2.3.4 Question 4 (Textbook)

A 60.0-kg person jumps onto the floor from a height of 3.00 m. If he lands stiffly, calculate the force on the knee joints.

2.4 Momentum

2.4.1 Question 1 (Textbook)

As all you guys do Maths Advanced and above, using your knowledge of calculus (outside scope of Physics syllabus), take the time derivative of momentum. What quantity do you get? i.e find $\frac{dp}{dt}$ where $p = mv$. Explain this conceptually by giving an example.

2.4.2 Question 2 (Textbook)

During the 2007 French Open, Venus Williams hit the fastest recorded serve in a premier women's match, reaching a speed of 58 m/s (209 km/h). What is the average force exerted

on the 0.057-kg tennis ball by Venus Williams' racquet, assuming that the ball's speed just after impact is 58 m/s, that the initial horizontal component of the velocity before impact is negligible, and that the ball remained in contact with the racquet for 5.0 ms (milliseconds)?

2.4.3 Question 3 (Textbook)

Using your answer from Question 1, explain the conservation of momentum from the lens of Newton's second law, what quantity/quantities are conserved?

2.4.4 Question 4 (Textbook)

Distinguish between elastic and inelastic collision. What is conserved in an elastic collision?

2.4.5 Question 5 (Textbook)

Calculate the velocities of two objects following an elastic collision, given that $m_1 = 0.5 \text{ kg}$, $m_2 = 3.5 \text{ kg}$, $v_1 = 4 \text{ ms}^{-1}$ and $v_2 = 0 \text{ ms}^{-1}$.

2.5 Projectile Motion

2.5.1 Question 1 (Jacaranda Q20)

A gymnast wants to jump a distance of 2.5 m, leaving the ground at an angle of 28° . With what speed must the gymnast take off?

2.5.2 Question 2 (Jacaranda Q21)

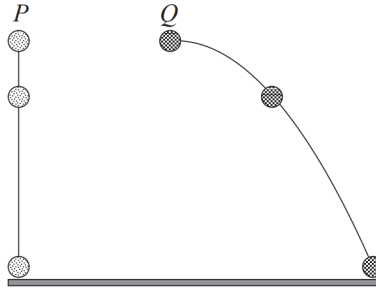
A horse rider wants to jump a 3.0 m wide stream. The horse can approach the stream with a speed of 7 ms^{-1} . At what angle must the horse take off?

2.5.3 Question 3 (Pearson Q2)

What is the angle that maximises horizontal distance? Show by taking a calculus approach.

2.5.4 Question 4 (HSC 2024 Q9)

- 9 Object P is dropped from rest, and object Q is launched horizontally from the same height.

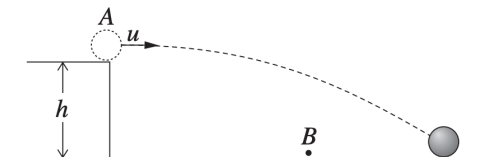


Which option correctly compares the projectile motion of P and Q ?

- A. The acceleration of P is less than the acceleration of Q .
- B. The final velocity of Q is greater than the final velocity of P .
- C. The time of flight of Q is greater than the time of flight of P .
- D. The initial vertical velocity of P is less than the initial vertical velocity of Q .

2.5.5 Question 5 (HSC 2023 Q8)

- 8 A ball is launched from a platform at position A with velocity u . It lands in the position shown.



The ball could be made to land at position B by increasing the

- A. velocity u .
- B. launch angle.
- C. mass of the ball.
- D. height of the platform.

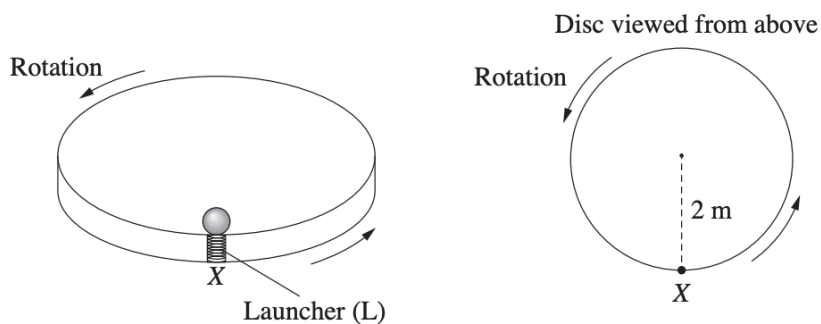
2.5.6 Question 6 (HSC 2023 Q32)

Question 32 (7 marks)

A horizontal disc rotates at 3 revolutions per second around its centre, with the top of the disc at ground level.

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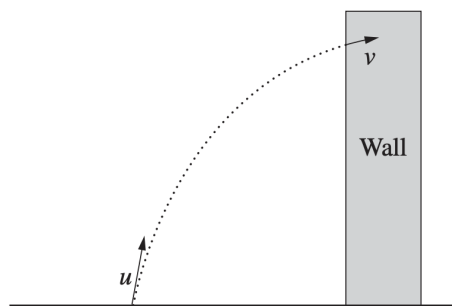
At 2 m from the centre of the disc, a ball is held in place at ground level on the top of the disc by a spring-loaded projectile launcher. At position X, the launcher fires the ball vertically upward with a velocity of 5.72 m s^{-1} .



Calculate the ball's position relative to the launcher's new position, at the instant the ball hits the ground.

2.5.7 Question 7 (HSC 2022 Q4)

8 An object is launched with an initial velocity, u , and hits a wall with a final velocity, v .

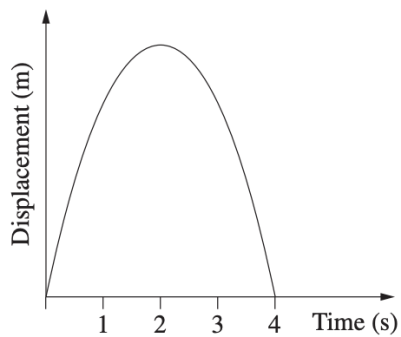


Which statement correctly compares components of u and v ?

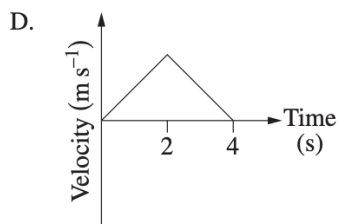
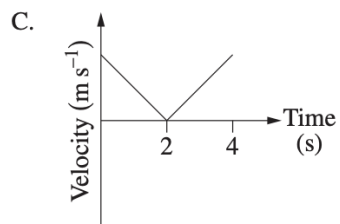
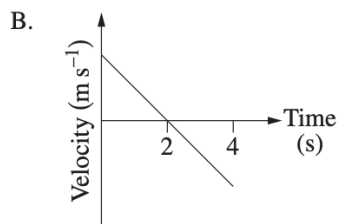
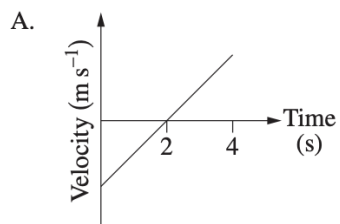
- A. The vertical component of v is less than the vertical component of u .
- B. The vertical component of v is greater than the vertical component of u .
- C. The horizontal component of v is less than the horizontal component of u .
- D. The horizontal component of v is greater than the horizontal component of u .

2.5.8 Question 8 (HSC 2022 Q11)

- 11 A projectile is launched vertically upwards. The displacement of the projectile as a function of time is shown.



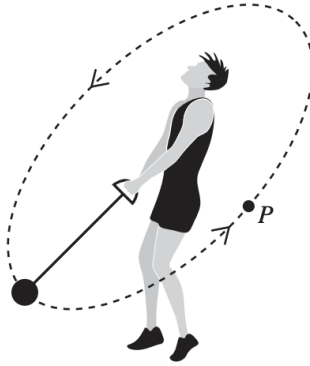
Which velocity–time graph corresponds to this motion?



2.5.9 Question 9 (HSC 2022 Q33)

Question 33 (6 marks)

In a hammer throw sport event, a 7.0 kg projectile rotates in a circle of radius 1.6 m, with a period of 0.50 s. It is released at point P , which is 1.2 m above the ground, where its velocity is at 45° to the horizontal.



- (a) Show that the vertical component of the projectile's velocity at P is 14.2 m s^{-1} . 2

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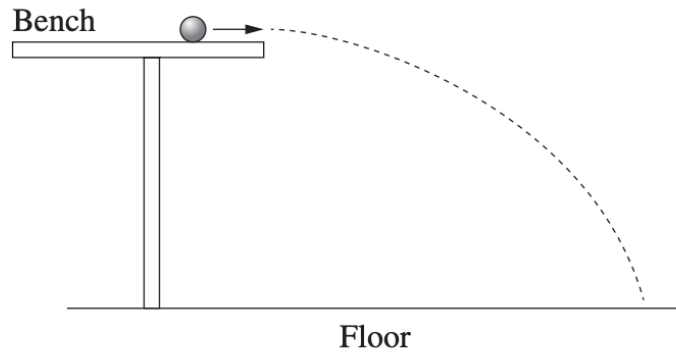
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- (b) Calculate the horizontal range of the projectile from point P . 4

2.5.10 Question 10 (HSC 2021 Q1)

- 1 A marble is rolled off a horizontal bench and falls to the floor.



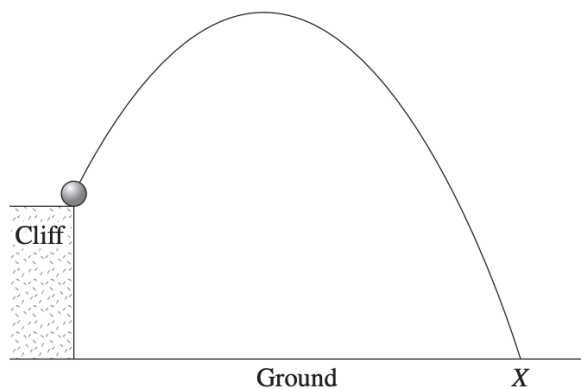
Rolling the marble at a slower speed would

- A. increase the range.
- B. decrease the range.
- C. increase the time of flight.
- D. decrease the time of flight.

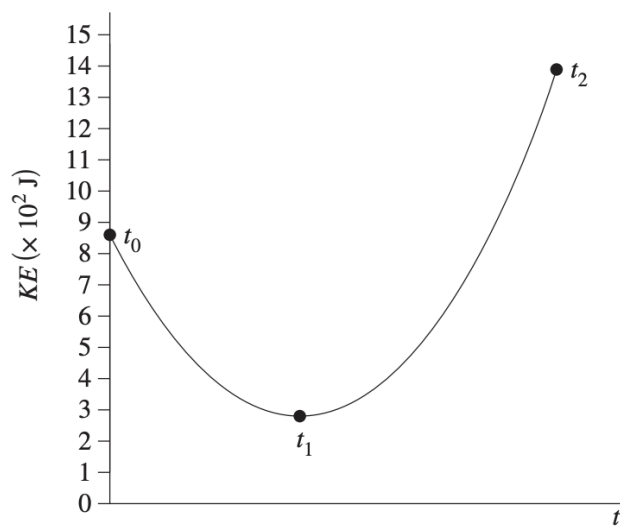
2.5.11 Question 11 (HSC 2021 Q34)

Question 34 (7 marks)

A 3.0 kg mass is launched from the edge of a cliff.



The kinetic energy of the mass is graphed from the moment it is launched until it hits the ground at X. The kinetic energy of the mass is provided for times t_0 , t_1 and t_2 .



<i>Time</i> (<i>t</i>)	<i>KE</i> (J)
t_0	864
t_1	284
t_2	1393

- (a) Account for the relative values of kinetic energy at t_0 , t_1 and t_2 . 4

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- (b) The horizontal component of the velocity of the mass during its flight is 13.76 ms^{-1} . 3

Calculate the time of flight of the mass.

2.5.12 Question 12 (HSC 2020 Q5)

- 5** A student throws a ball that follows a parabolic trajectory.

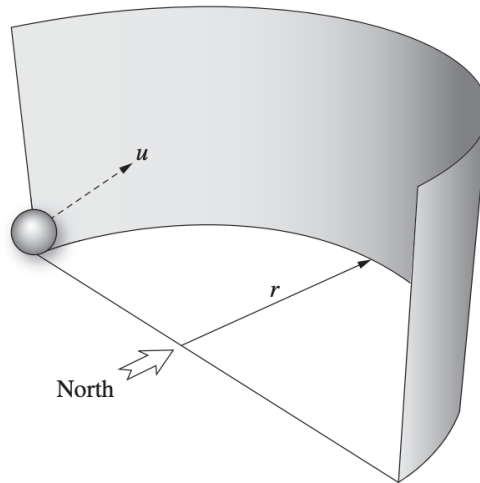
What change to the initial velocity would make the ball's time of flight shorter?

- A. Increasing only the vertical component
- B. Decreasing only the vertical component
- C. Increasing only the horizontal component
- D. Decreasing only the horizontal component

2.5.13 Question 13 (HSC 2020 Q20)

20 The diagram shows a smooth, semi-circular, vertical wall with radius, r .

A ball is launched from the position shown with a velocity u towards north at an angle to the horizontal.



The ball follows a trajectory around the wall before landing on the ground, opposite its starting point. It does not reach the top of the wall.

Assume that there is no friction between the ball and the wall.

Which statement correctly describes the net force acting on the ball during its motion?

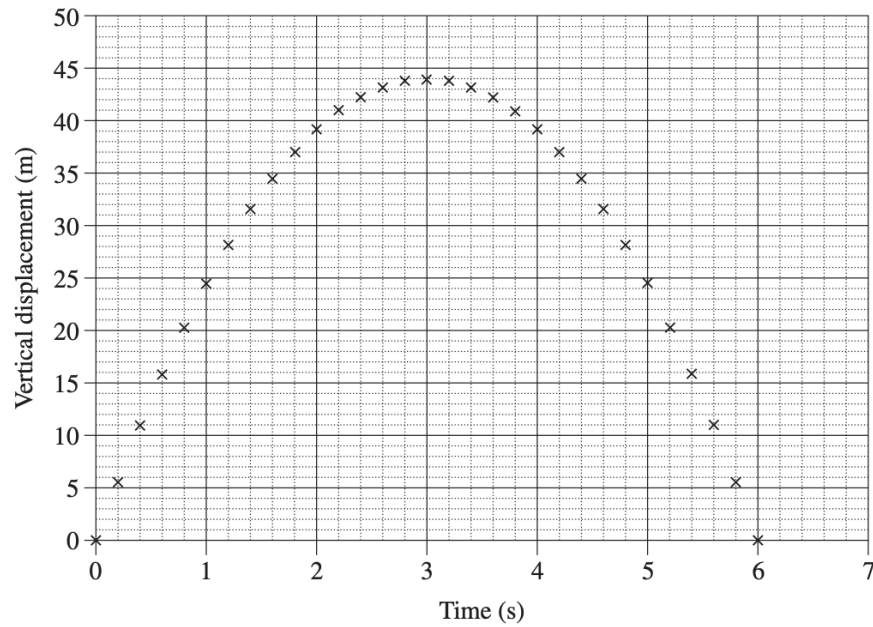
- A. The magnitude of the net force remains constant.
- B. The direction of the net force is vertically downwards.
- C. The direction of the net force is perpendicular to the wall.
- D. The magnitude of the net force reaches a minimum when the ball is at its highest point.

2.5.14 Question 14 (HSC 2020 Q24)

Question 24 (4 marks)

The graph shows the vertical displacement of a projectile throughout its trajectory. The range of the projectile is 130 m.

4

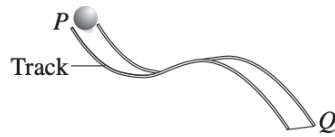


Calculate the initial velocity of the projectile.

2.5.15 Question 15 (HSC 2019 Q30)

Question 30 (6 marks)

A ball, initially at rest in position P , travels along a frictionless track to point Q and then falls to strike the floor below.



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At the instant the ball leaves the track at Q it has a velocity of 1.5 m s^{-1} at an angle of 50° to the horizontal.

- (a) Calculate the difference in height between P and Q .

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- (b) The ball takes 0.5 s to reach the floor after leaving the track at Q .

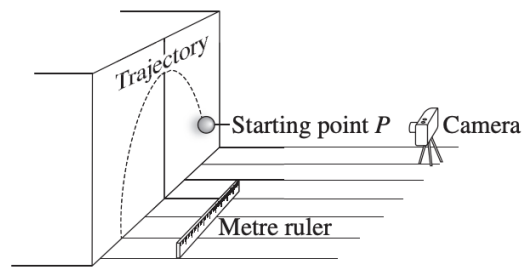
3

Calculate the height of Q above the floor.

2.5.16 Question 16 (HSC 2018 Q27)

Question 27 (6 marks)

- (a) The diagram shows a camera and a ruler set up to obtain data about a projectile's motion along the trajectory shown. The entire trajectory is visible through the camera. 3

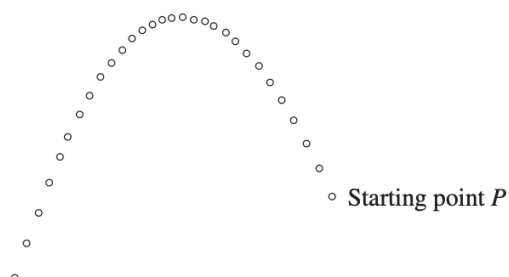


Identify ONE of the errors in this set-up and describe the effect of this error on the results.

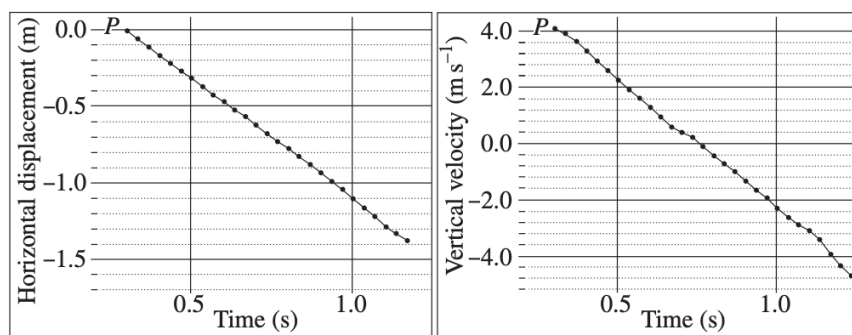
Question 27 (continued)

- (b) An experiment was set up based on the method described in part (a), but conducted so that the data obtained were valid. 3

The image shows the trajectory of the ball.



The graphs show data from this experiment.

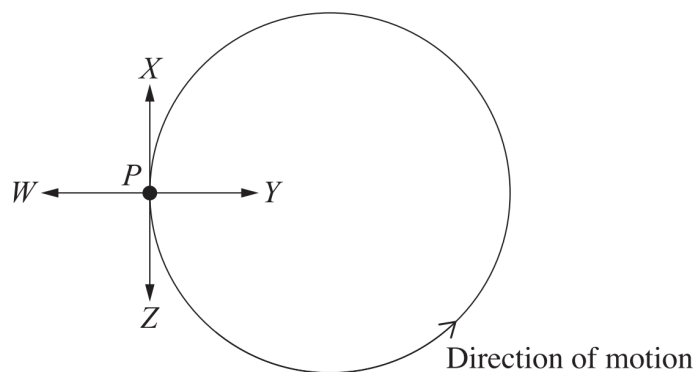


Using the graphs, describe the velocity and acceleration of the ball quantitatively and qualitatively.

2.6 Circular Motion

2.6.1 Question 1 (HSC 2024 Q1)

- 1 The diagram shows an object, P , undergoing uniform circular motion.



Which arrow shows the direction of the net force acting on P ?

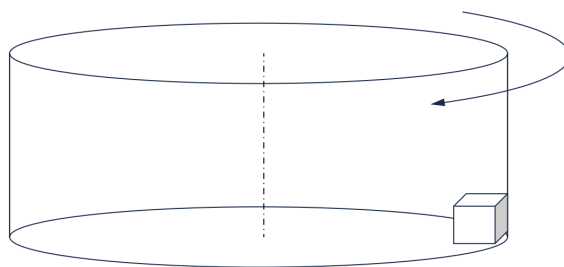
- A. W
- B. X
- C. Y
- D. Z

2.6.2 Question 2 (HSC 2024 Q30)

Question 30 (4 marks)

An object sits on the floor of a hollow cylinder rotating around an axis, as shown. The cylinder's rotation causes the object to undergo uniform circular motion.

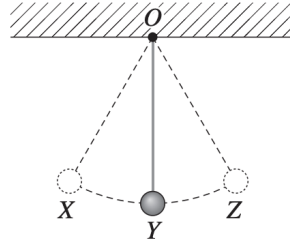
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Explain the effect on all of the forces acting on the object if the period of the cylinder's rotation is halved. Ignore the effects of friction.

2.6.3 Question 3 (HSC 2023 Q17)

- 17 A mass attached to a lightweight, rigid arm hanging from point O , oscillates freely between X and Z .



Which statement best describes the torque acting on the arm as it oscillates?

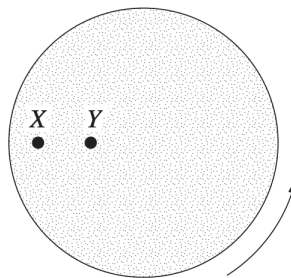
- A. It is constant in magnitude and direction.
- B. It is zero at Y and a maximum at X and Z .
- C. It is zero at X and Z and a maximum at Y .
- D. It is constant in magnitude but its direction changes.

2.6.4 Question 4 (HSC 2021 Q22)

Question 22 (3 marks)

A horizontal disc rotates at a constant rate as shown. Two points on the disc, X and Y , are labelled. X is twice as far away from the centre of the disc as Y .

3

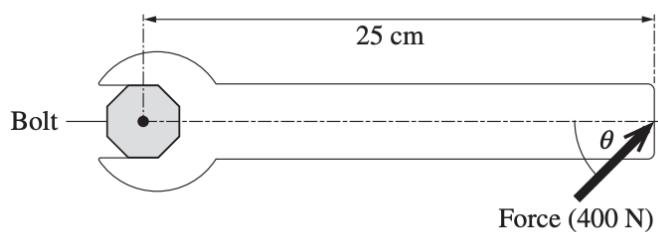


Compare the angular and instantaneous velocities of X with those of Y .

2.6.5 Question 5 (HSC 2019 Q26)

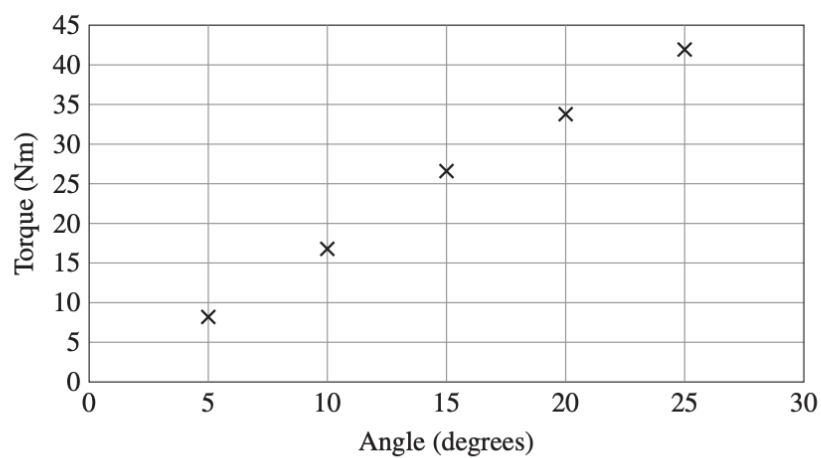
Question 26 (6 marks)

A student carried out an experiment to investigate the relationship between the torque produced by a force and the angle at which the force is applied. A 400 N force was applied to the same position on the handle of a spanner at different angles, as shown.



A high-precision device measured the torque applied to the bolt.

The data from the experiment is graphed below.



Question 26 (continued)

The student concluded that the torque (τ) was proportional to the angle (θ) and proposed the model

$$\tau = k\theta$$

where $k = 1.7 \text{ Nm/degree}$.

- (a) Justify the validity of the student's model using information from the graph. **3**

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- (b) What happens to the accuracy of this model's predictions as the angle increases beyond 25° ? Justify your answer with reference to a different model. **3**

2.7 Gravitational Fields of planets

2.7.1 Question 1 (Textbook)

What are Kepler's three laws of motion. Describe the various experiments or observations that verified his ideas.

2.7.2 Question 2 (Textbook)

Distinguish between a geostationary and a low earth orbit (LEO). What are the benefits and costs of each orbit?

2.7.3 Question 3 (Textbook)

Explain how a Hohmann transfer orbit works. State the energy changes at each significant point.

2.7.4 Question 4 (HSC 2024 Q11)

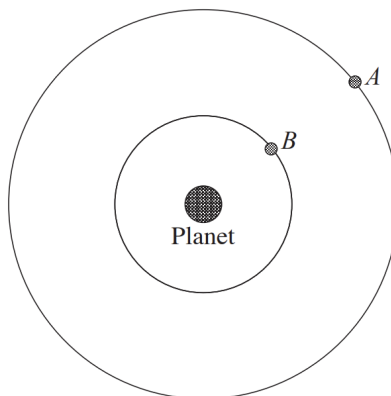
- 11** A satellite is in a circular orbit.

What is the relationship between its orbital velocity, v , and its orbital radius, r ?

- A. v is directly proportional to the square of r .
- B. v is inversely proportional to the square of r .
- C. v is directly proportional to the square root of r .
- D. v is inversely proportional to the square root of r .

2.7.5 Question 5 (HSC 2024 Q13)

- 13 The diagram shows two identical satellites, A and B , orbiting a planet.



Which row in the table correctly compares the potential energy, U , and kinetic energy, K , of the satellites?

	Potential energy	Kinetic energy
A.	$U_A > U_B$	$K_A < K_B$
B.	$U_A < U_B$	$K_A > K_B$
C.	$U_A > U_B$	$K_A > K_B$
D.	$U_A < U_B$	$K_A < K_B$

2.7.6 Question 6 (HSC 2024 Q25)

Question 25 (6 marks)

The mathematical model below shows the relationship between the orbital radius of a satellite and its period.

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

- (a) By considering gravitational force, show how this model can be derived.

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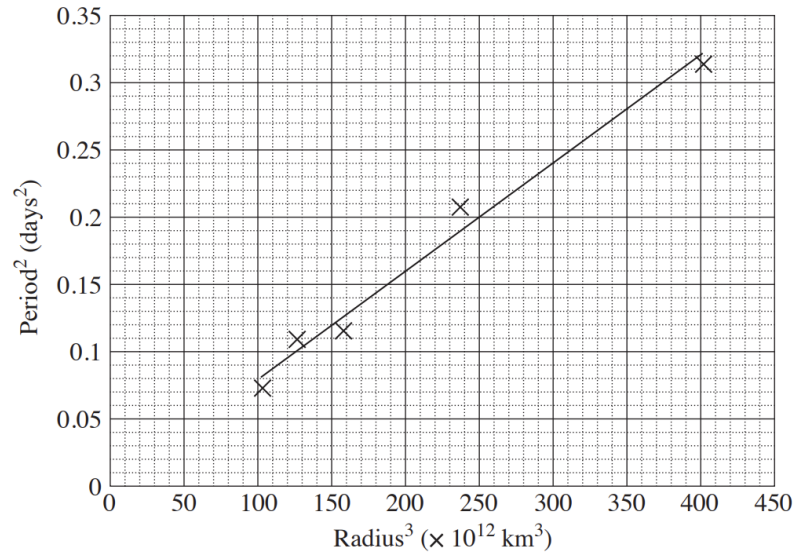
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Question 25 (continued)

- (b) A planet with five moons is discovered. The following graph is produced from observations of the orbital radius of the moons and their orbital periods, measured in Earth days.

4



Use the graph to calculate the mass of the planet.

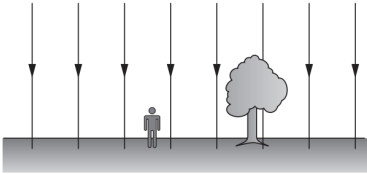
2.7.7 Question 7 (HSC 2024 Q6)

Question 31 (4 marks)

In a thought experiment, a projectile is launched vertically from Earth's surface. Its initial velocity is less than the escape velocity.

4

The behaviour of the projectile can be analysed by using two different models, Model *A* and Model *B* as shown.

Model <i>A</i>	Model <i>B</i>
	
Earth's gravitational field is uniform	Earth's gravitational field is radial

The effects of Earth's atmosphere and Earth's rotational and orbital motions can be ignored.

Compare the maximum height reached by the projectile, using each model. In your answer, describe the energy changes of the projectile.

2.7.8 Question 8 (HSC 2023 Q1)

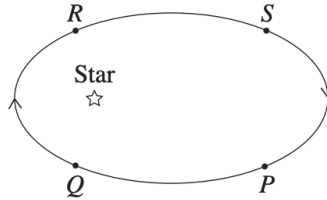
- 1 The gravitational field strength acting on a spacecraft decreases as its altitude increases.

This is due to a change in the

- A. mass of Earth.
- B. mass of the spacecraft.
- C. density of the atmosphere.
- D. distance of the spacecraft from Earth's centre.

2.7.9 Question 9 (HSC 2023 Q5)

- 5 An exoplanet is in an elliptical orbit, moving in the direction shown. The distances between consecutive positions P , Q , R and S are equal.



Between which two points is the exoplanet's travel time the greatest?

- A. P and Q
- B. Q and R
- C. R and S
- D. S and P

2.7.10 Question 10 (HSC 2023 Q14)

- 14 Planet X has a mass 4 times that of Earth and a radius 3 times that of Earth. The escape velocity at the surface of Earth is 11.2 km s^{-1} .

What is the escape velocity at the surface of planet X ?

- A. 8.40 km s^{-1}
- B. 9.70 km s^{-1}
- C. 12.9 km s^{-1}
- D. 14.9 km s^{-1}

2.7.11 Question 11 (HSC 2023 Q23)

Question 23 (7 marks)

The James Webb Space Telescope (JWST) has a mass of 6.1×10^3 kg and orbits the Sun at a distance of approximately 1.52×10^{11} m.

- (a) The Sun has a mass of 1.99×10^{30} kg. **2**

Calculate the magnitude of gravitational force the Sun exerts on the JWST.

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- (b) The telescope is sensitive to wavelengths from 6.0×10^{-7} m to 2.8×10^{-5} m. **3**

What is the minimum photon energy that it can detect?

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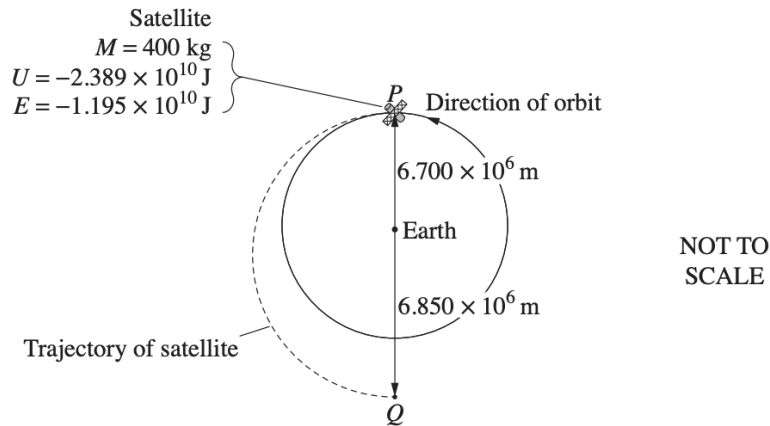
- (c) The JWST observed an exoplanet emitting a peak wavelength of 1.14×10^{-5} m. **2**

Calculate the temperature of the exoplanet.

2.7.12 Question 12 (HSC 2023 Q34)

Question 34 (9 marks)

A 400 kg satellite is travelling in a circular orbit of radius 6.700×10^6 m around Earth. Its potential energy is -2.389×10^{10} J and its total energy is -1.195×10^{10} J.



At point P , the satellite's engines are fired, increasing the satellite's velocity in the direction of travel and causing its kinetic energy to increase by 5.232×10^8 J. Assume that this happens instantaneously and that the engine is then shut down.

The satellite follows the trajectory shown, which passes through Q , 6.850×10^6 m from Earth's centre.

- (a) Analyse qualitatively the energy changes as the satellite moves from P to Q .

2

- (b) Show that the kinetic energy of the satellite at Q is $1.194 \times 10^{10} \text{ J}$. 4

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- (c) Explain the motion of the satellite after it passes through Q . 3

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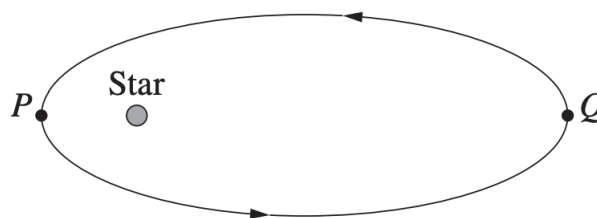
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2.7.13 Question 13 (HSC 2022 Q6)

- 6 The elliptical orbit of a planet around a star is shown.



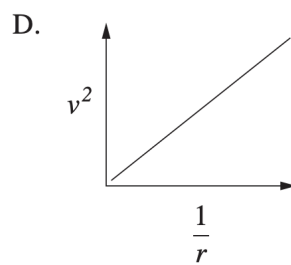
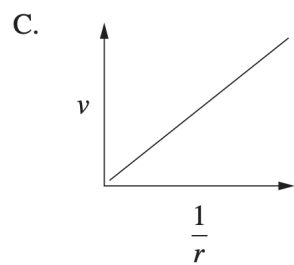
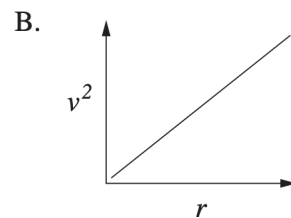
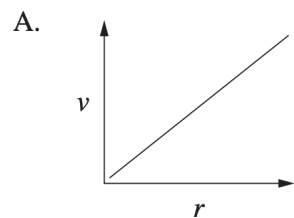
Which type of energy is greater at position P than at Q ?

- A. Kinetic
- B. Nuclear
- C. Potential
- D. Total

2.7.14 Question 14 (HSC 2022 Q10)

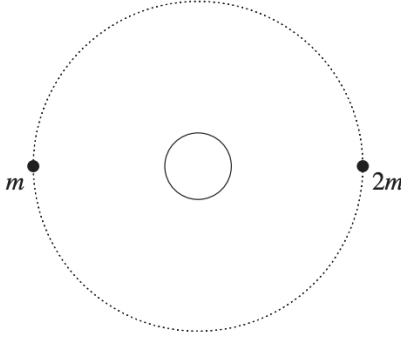
- 10 The orbital velocity, v , of a satellite around a planet is given by $v = \sqrt{\frac{GM}{r}}$.

Which graph is consistent with this relationship?



2.7.15 Question 15 (HSC 2022 Q13)

- 13 Two satellites share an orbit around a planet. One satellite has twice the mass of the other.



Which quantity would be different for the two satellites?

- A. Speed
- B. Momentum
- C. Orbital period
- D. Centripetal acceleration

2.7.16 Question 16 (HSC 2022 Q25)

Question 25 (5 marks)

A rocket is launched vertically from a planet of mass M . After it leaves the atmosphere, the rocket's engine is turned off and it continues to move away from the planet. From this time the rocket's mass is 200 kg. The rocket's speed, v , at two different distances from the planet's centre, R , is shown.

Point	R (m)	v (m s^{-1})
1	4.3×10^6	5500
2	2.5×10^7	2900

- (a) Show that the magnitude of the change in kinetic energy from point 1 to point 2 is 2.2×10^9 J. 2

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- (b) Determine the mass M of the planet using the law of conservation of energy. 3

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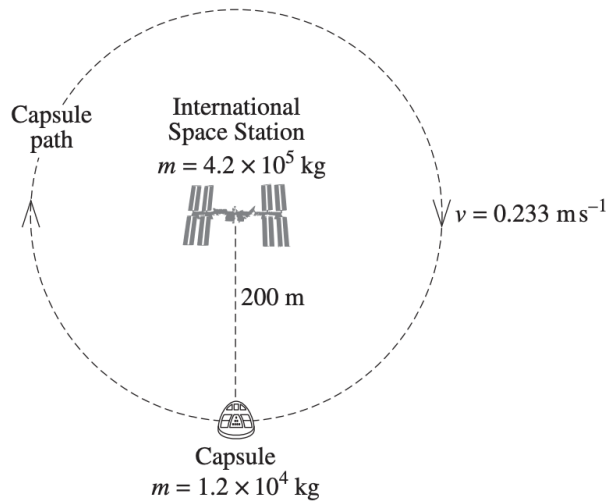
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2.7.17 Question 17 (HSC 2022 Q35)

Question 35 (5 marks)

A capsule travels around the International Space Station (ISS) in a circular path of radius 200 m as shown.

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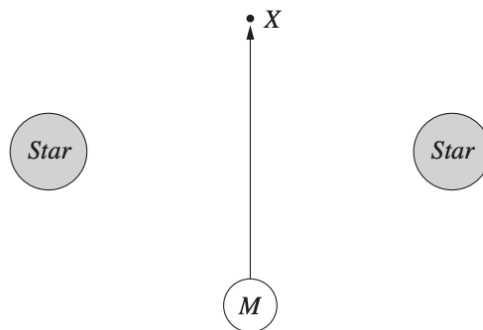
Analyse this system to test the hypothesis below.

The uniform circular motion of the capsule around the ISS can be accounted for in terms of the gravitational force between the capsule and the ISS.

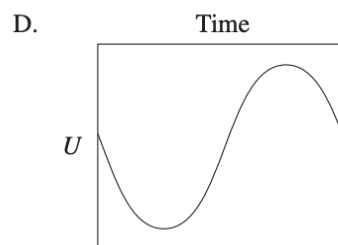
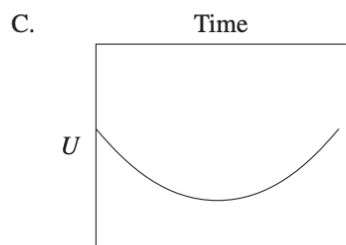
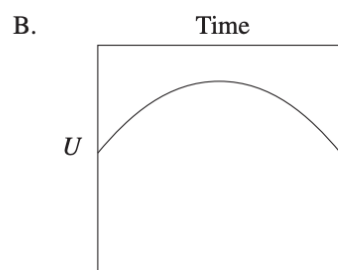
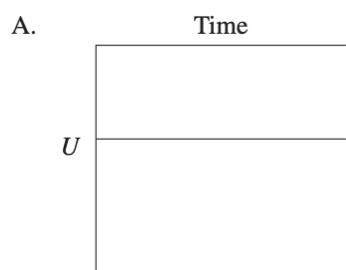
2.7.18 Question 18 (HSC 2021 Q1)

- 9 A mass, M , is positioned at an equal distance from two identical stars as shown.

The mass is then moved to position X .



Which graph best represents the gravitational potential energy, U , of the mass during this movement?



2.7.19 Question 19 (HSC 2021 Q25)

Question 25 (5 marks)

A satellite is launched from the surface of Mars into an orbit that keeps it directly above a position on the surface of Mars.

Mass of Mars = 6.39×10^{23} kg

Length of Martian day = 24 hours and 40 minutes

- (a) Identify TWO energy changes as the satellite moves from the surface of Mars into orbit. **2**

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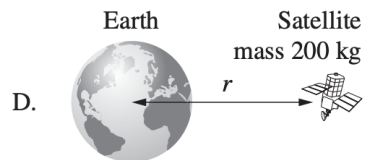
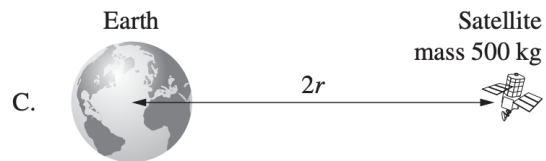
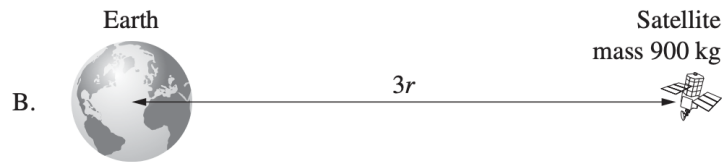
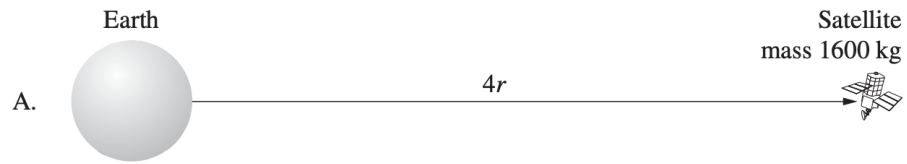
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- (b) Calculate the orbital radius of the satellite. **3**

2.7.20 Question 20 (HSC 2020 Q12)

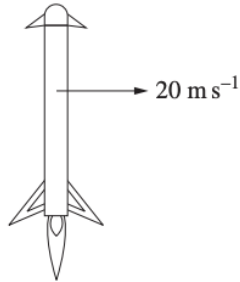
- 12 In which of the following would the satellite have the greatest escape velocity from Earth?



2.7.21 Question 21 (HSC 2020 Q15)

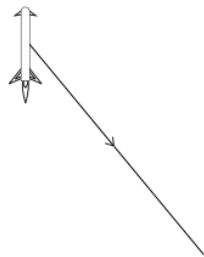
- 15 A rocket returns to Earth for reuse after launching satellites, using its engines to make a controlled landing.

The rocket having a mass of 7800 kg is on approach to the ground, travelling horizontally at 20 m s^{-1} as shown in the diagram, when the engine thrust is changed to 90 000 newtons.

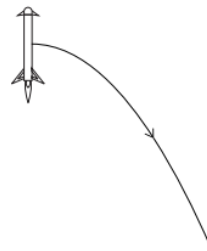


Which diagram shows the trajectory of the rocket following this change of thrust?

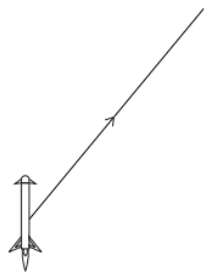
A.



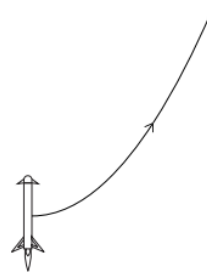
B.



C.

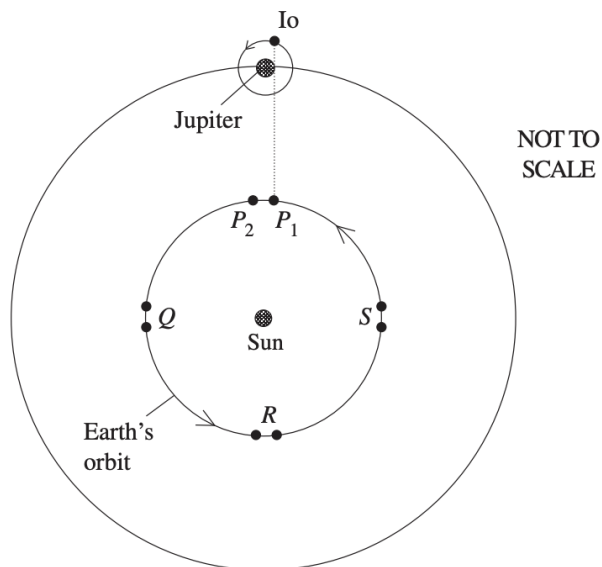


D.



2.7.22 Question 22 (HSC 2020 Q18)

- 18 An observer sees Io complete one orbit of Jupiter as Earth moves from P_1 to P_2 , and records the observed orbital period as t_P . Similarly, the time for one orbit of Io around Jupiter was measured as Earth moved between the pairs of points at Q, R and S , with the corresponding measured periods of Io being t_Q , t_R and t_S .



Which measurement of the orbital period would be the longest?

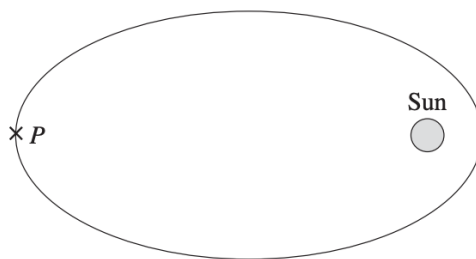
- A. t_P
- B. t_Q
- C. t_R
- D. t_S

2.7.23 Question 23 (HSC 2020 Q31)

Question 31 (6 marks)

- (a) The orbit of a comet is shown.

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Account for the changes in velocity of the comet as it completes one orbit from position P .

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- (b) Two stars, A and B , of equal mass m , separated by a distance x , interact gravitationally such that the speed of A is constant.

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Derive an expression for the speed of B .

2.7.24 Question 24 (HSC 2019 Q9)

- 9 Two satellites have the same mass. One (LEO) is in low-Earth orbit and the other (GEO) is in a geostationary orbit.

The total energy of a satellite is half its gravitational potential energy.

Which row of the table correctly identifies the satellite with the greater orbital period and the satellite with the greater total energy?

	<i>Greater orbital period</i>	<i>Greater total energy</i>
A.	LEO	LEO
B.	LEO	GEO
C.	GEO	LEO
D.	GEO	GEO

2.7.25 Question 25 (HSC 2019 Q14)

- 14 A satellite in circular orbit at a distance r from the centre of Earth has an orbital velocity v .

If the distance was increased to $2r$, what would be the satellite's orbital velocity?

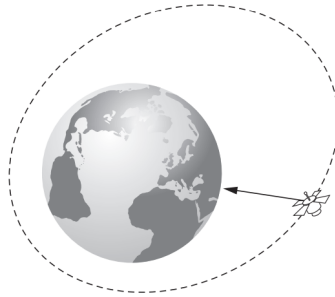
- A. $\frac{v}{2}$
B. $0.7v$
C. $1.4v$
D. $2v$

2.7.26 Question 26 (HSC 2018 Q1)

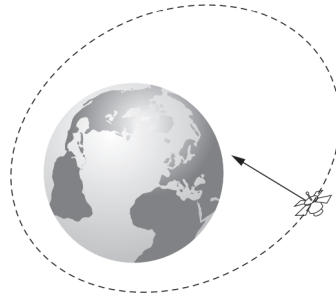
- 1 A satellite orbits Earth as shown.

Which diagram correctly shows the direction of the satellite's acceleration?

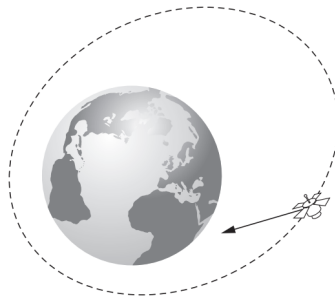
A.



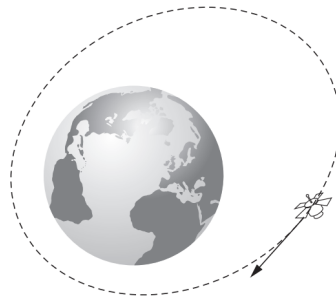
B.



C.



D.



2.7.27 Question 27 (HSC 2018 Q7)

- 7 A planet X has twice the mass and twice the radius of Earth.

What is the magnitude of the gravitational acceleration close to the surface of planet X?

- A. $\frac{1}{2}g$
B. $1g$
C. $2g$
D. $4g$

2.7.28 Question 28 (HSC 2018 Q28)

Question 28 (5 marks)

The radius of the moon is 1740 km. The moon's mass is 7.35×10^{22} kg. In this question, ignore the moon's rotational and orbital motion.

A 20 kg mass is launched vertically from the moon's surface at a velocity of 1200 m s^{-1} .

- (a) Show that the change in potential energy of the mass in moving from the surface to an altitude of 500 km is $1.26 \times 10^7 \text{ J}$. 2

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- (b) Calculate the velocity of the 20 kg mass at an altitude of 500 km. 3

3 Electromagnetism

3.1 Electrical Fields, Potential difference and Work

3.1.1 Question 1 (Textbook)

What is the difference between the electric field strength and the potential difference, explain using relevant questions. Explain these concepts to a 5 year old.

3.1.2 Question 2 (Textbook)

What do I mean when I say a battery does work on each particle in the circuit? How do you calculate the total work done on the circuit by the battery, if I hooked up a load to the circuit, what would this work do? Give an example.

3.1.3 Question 3 (Jacaranda Q3)

Calculate the acceleration of an electron in a uniform electric field strength of $1 \times 10^6 \text{ NC}^{-1}$. How long would it take the electron to reach 10% speed of light, ignoring relativistic effects, what is the distance the electron covered. Assuming that this distance is the separation of two potentials, what is the potential difference?

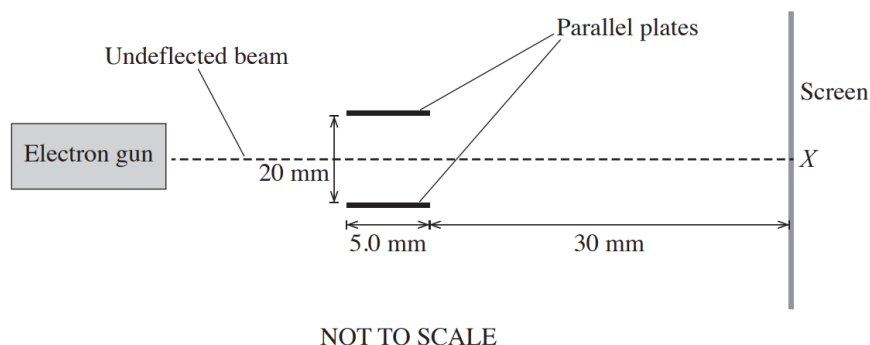
3.1.4 Question 4 (HSC 2024 Q28)

Question 28 (7 marks)

An electron gun fires a beam of electrons at $2.0 \times 10^6 \text{ m s}^{-1}$ through a pair of parallel charged plates towards a screen that is 30 mm from the end of the plates as shown.

There is a uniform electric field between the plates of $1.5 \times 10^4 \text{ N C}^{-1}$. The plates are 5.0 mm wide and 20 mm apart. The electron beam enters mid-way between the plates. X marks the spot on the screen where an undeflected beam would strike.

Ignore gravitational effects on the electron beam.



- (a) Show that the acceleration of an electron between the parallel plates is $2.6 \times 10^{15} \text{ m s}^{-2}$. **2**
- (b) Show that the vertical displacement of the electron beam at the end of the parallel plates is approximately 8.1 mm. **2**

Question 28 (continued)

- (c) How far from point X will the electron beam strike the screen? **3**

3.1.5 Question 5 (HSC 2023 Q12)

- 12 Figure I shows a positively charged particle accelerating freely from X to Y , between oppositely charged plates. The change in the particle's kinetic energy is W .

The distance between the plates is then doubled as shown in Figure II. The same charge accelerates from rest over the same distance from X to Y .

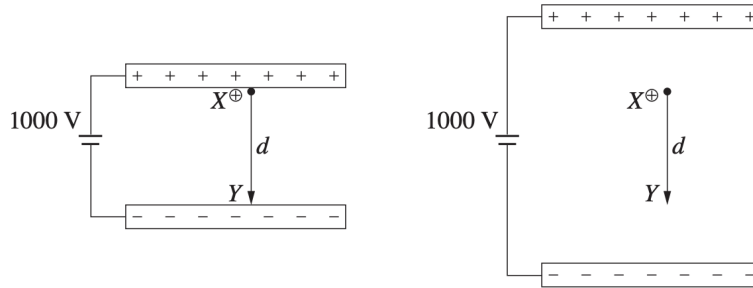


Figure I

Figure II

What is the change in kinetic energy of the positively charged particle shown in Figure II?

- A. W
- B. $\frac{W}{2}$
- C. $\sqrt{W}$
- D. $2W$

3.1.6 Question 6 (HSC 2023 Q18)

- 18** The diagrams show the trajectories of two particles with the same mass and charge and which initially have the same velocity u , as shown. The subsequent motion of each particle is determined by its properties and by its interaction with the field in which it is moving.

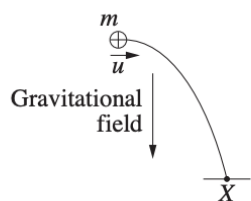


Figure I

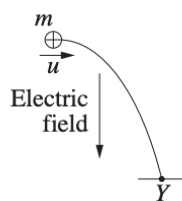


Figure II

X and Y represent the landing points in Figures I and II.

Which row of the table shows the correct paths of the particles if the mass of each is increased by the same amount and they are given the same initial velocity u ?

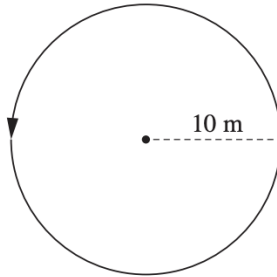
	Gravitational field	Electric field
A.		
B.		
C.		
D.		

3.1.7 Question 7 (HSC 2023 Q24)

Question 24 (3 marks)

An electron is travelling at $3.0 \times 10^6 \text{ m s}^{-1}$ in the path shown.

3

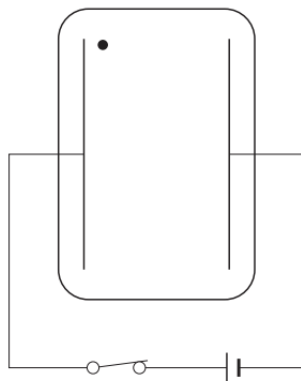


Calculate the magnetic field required to keep the electron in the path.

3.1.8 Question 8 (HSC 2021 Q18)

- 18 An evacuated chamber contains a pair of parallel plates connected to a power supply and a switch which is initially closed.

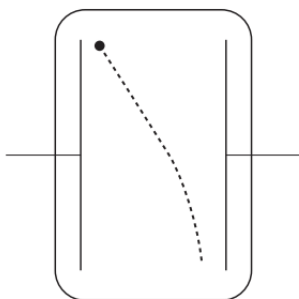
A positively charged mass (●) falls within the chamber, under the influence of gravity, from the position shown.



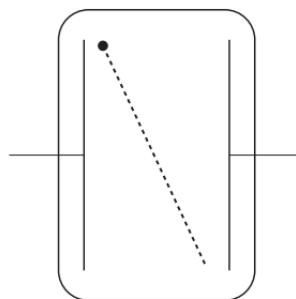
When the mass has fallen half the height of the chamber, the switch is opened.

Which of the following correctly shows the trajectory of the mass?

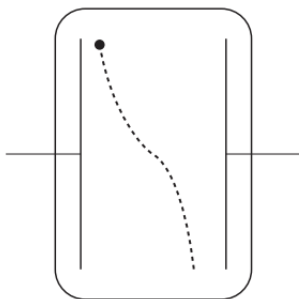
A.



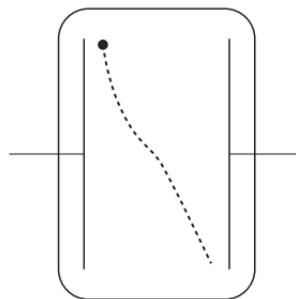
B.



C.



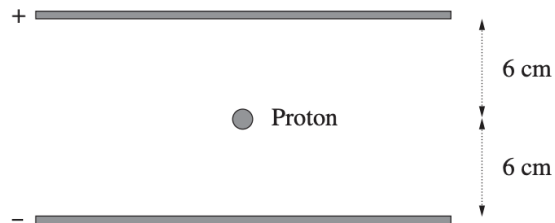
D.



3.1.9 Question 9 (HSC 2021 Q30)

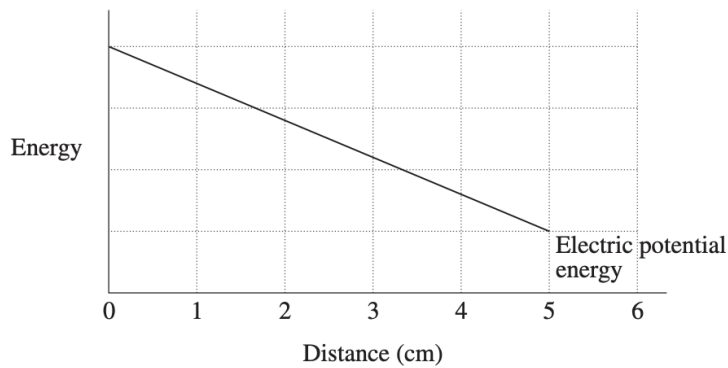
Question 30 (5 marks)

In an experiment, a proton accelerates from rest between parallel charged plates. The spacing of the plates is 12 cm and the proton is initially positioned at an equal distance from both plates, as shown. Ignore the effect of gravity.



- (a) The electrical potential energy of the proton is recorded in the following graph for the first 5 cm of its motion. 2

On the graph, sketch the corresponding kinetic energy of the proton over the same distance.

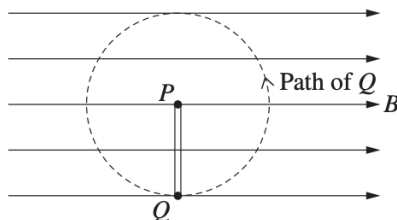


- (b) The experiment is repeated using an electron in the place of the proton. 3

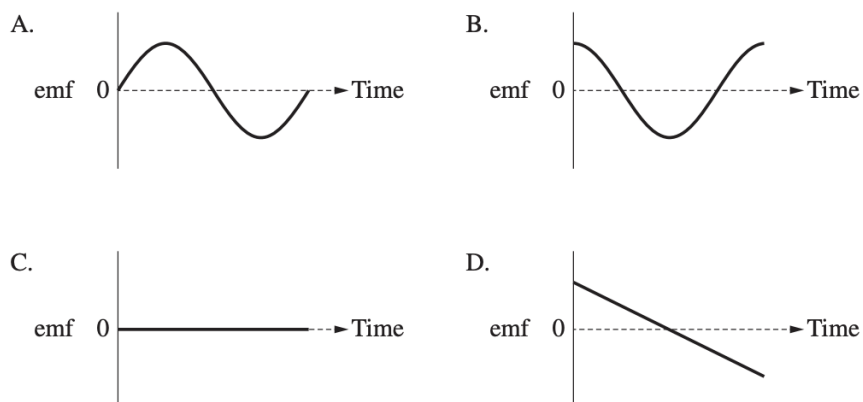
Explain how the motion of the electron would differ from that of the proton.

3.1.10 Question 10 (HSC 2020 Q1)

- 19 A conductor PQ is in a uniform magnetic field. The conductor rotates around the end P at a constant angular velocity.



Which graph shows the induced emf between P and Q as the conductor completes one revolution from the position shown?



3.1.11 Question 11 (HSC 2019 Q29)

Question 29 (3 marks)

A particle having mass m and charge q is accelerated from rest through a potential difference V . Assume that the only force acting on the particle is due to the electric field associated with this potential difference.

3

Show that the final velocity of the particle is given by $v = \sqrt{\frac{2qV}{m}}$.

3.1.12 Question 12 (HSC 2018 Q26)

Question 26 (4 marks)

Outline the similarities and differences between the effects of electric fields and gravitational fields on matter. In your answer, refer to the definitions of these fields.

4

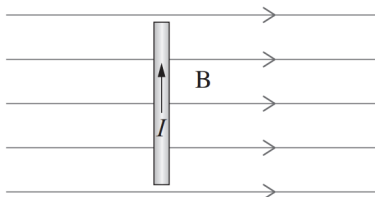
3.2 Magnetic Fields

3.2.1 Question 1 (Textbook)

Explain why magnetic fields do no work. Can you think of an example where you have encountered this kind of force before? What are the similarities? What is the physical significance, in terms of the particle's kinetic energy of no work being done on it by the magnetic field.

3.2.2 Question 1 (HSC 2024 Q10)

- 10 A rod carrying a current, I , placed in a uniform magnetic field as shown, experiences a force F .



How many degrees must the rod be rotated clockwise so that it experiences a force $\frac{F}{2}$?

- A. 30°
- B. 45°
- C. 60°
- D. 90°

3.2.3 Question 2 (HSC 2024 Q19)

- 19 In a vacuum chamber there is a uniform electric field and a uniform magnetic field.

A proton having a velocity, v , enters the chamber. Its velocity remains unchanged as it travels through the chamber.

A second proton having a velocity, $2v$, in the same direction as the first proton, then enters the chamber at the same point as the first proton.

In the chamber, the acceleration of the second proton

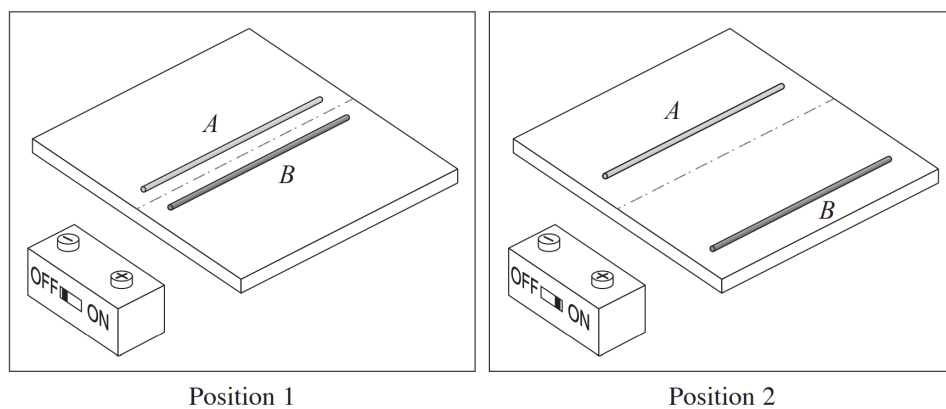
- A. is zero.
- B. is constant in magnitude and direction.
- C. changes in both magnitude and direction.
- D. is constant in magnitude, but not direction.

3.2.4 Question 3 (HSC 2024 Q29)

Question 29 (6 marks)

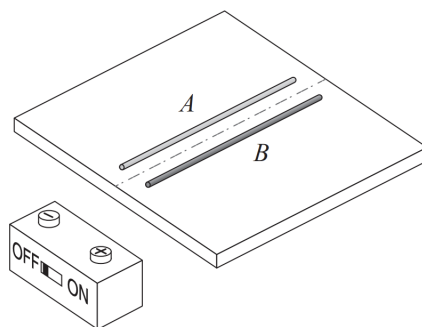
Two horizontal metal rods, A and B , of different materials are resting on a frictionless table. Initially they are at rest in position 1.

Both rods are then connected to a battery using wires. After the switch is turned on, currents of different magnitude flow in each rod. The rods move to position 2 after time, t . In position 2, B has a larger displacement than A from position 1. The masses of the wires are negligible.



- (a) Position 1 is reproduced below. Draw wires to show how the battery must be connected to the ends of the two rods in order for the magnitude of the current in each rod to be different, and for position 2 to be reached. No components, other than the wires, are required.

2



Question 29 (continued)

- (b) When the switch is turned on, the current in rod A is greater than the current in rod B .

4

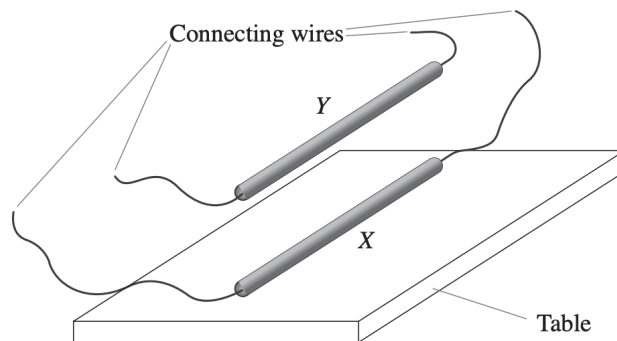
Consider this statement.

Position 2 results from the larger current in rod A , causing a larger force to act on rod B .

Evaluate this statement with reference to relevant physics principles.

3.2.5 Question 4 (HSC 2023 Q16)

- 16 In a thought experiment, two identical parallel aluminium rods, X and Y , are carrying electric currents of equal magnitude. Rod X rests on a table. Rod Y remains stationary, vertically above X , as a result of the magnetic interaction. The masses of the connecting wires are negligible.

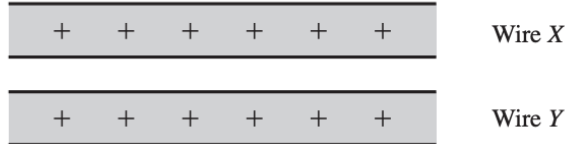


Which statement must be correct if rod Y is stationary?

- A. The magnetic force acting on X is upward.
- B. The currents through X and Y are in the same direction.
- C. The force the table exerts on X is equal and opposite to the total weight of X and Y .
- D. The force the table exerts on X is equal and opposite to the force of gravity acting on Y .

3.2.6 Question 5 (HSC 2023 Q19)

- 19 The diagram represents the distribution of positive charges in identical wires when no current is flowing.



Equal currents then flow in each wire, but in opposite directions. These currents are considered conventionally as the flow of positive charge.

Which diagram represents the charge distribution in the wires, from the frame of reference of a positive charge in wire Y?

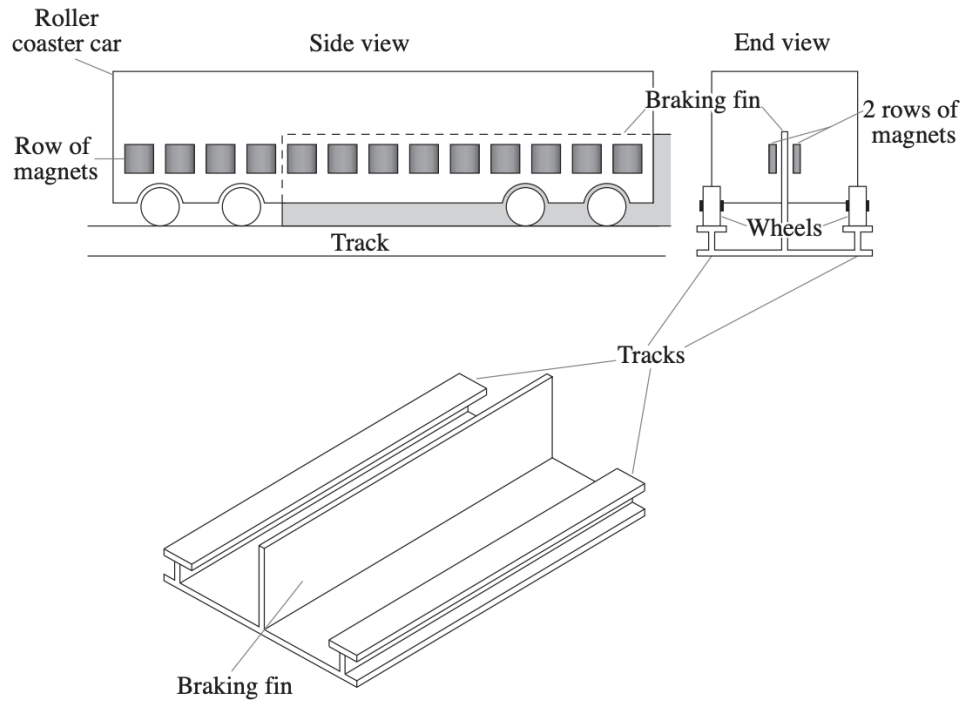
- A.
-
- Wire X
- Wire Y
- B.
-
- Wire X
- Wire Y
- C.
-
- Wire X
- Wire Y
- D.
-
- Wire X
- Wire Y

3.2.7 Question 6 (HSC 2023 Q31)

Question 31 (5 marks)

A roller coaster uses a braking system represented by the diagrams.

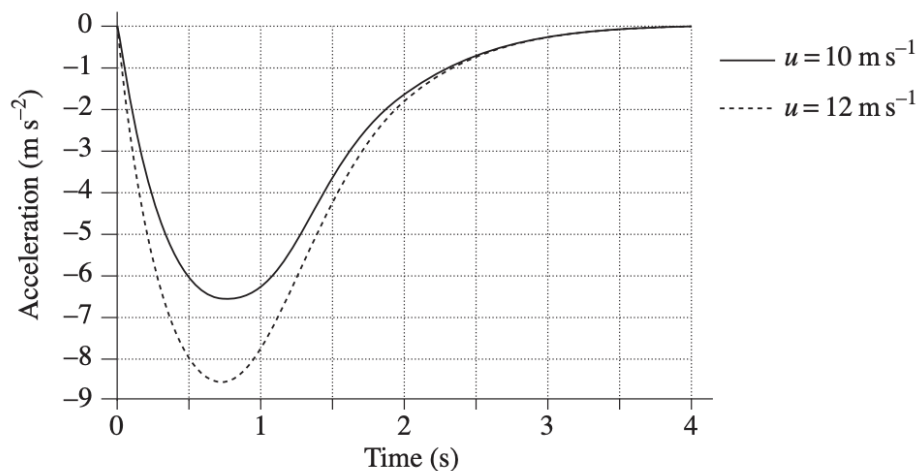
5



When the roller coaster car reaches the end of the ride, the two rows of permanent magnets on the car pass on either side of a thick aluminium conductor called a braking fin.

Question 31 (continued)

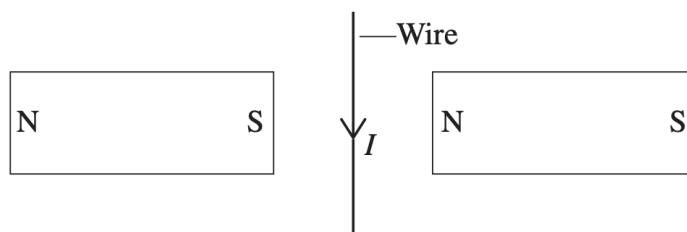
The graph shows the acceleration of the roller coaster reaching the braking fin at two different speeds.



Explain the similarities and differences between these two sets of data.

3.2.8 Question 7 (HSC 2022 Q4)

- 4 A current-carrying wire is in a magnetic field, as shown.

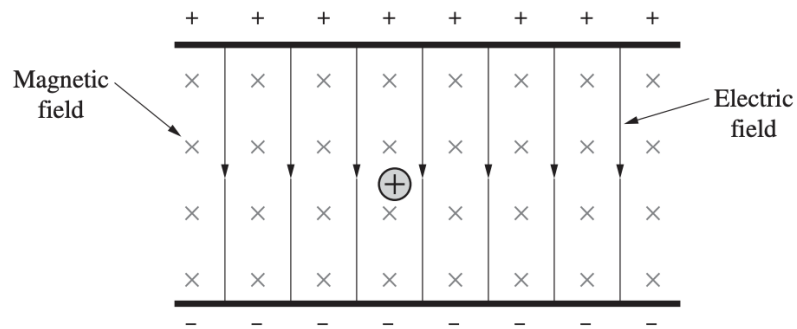


What is the direction of the force on the wire?

- A. Left
- B. Right
- C. Into the page
- D. Out of the page

3.2.9 Question 8 (HSC 2022 Q12)

- 12 The diagram shows a region in which there are uniform electric and magnetic fields. A positively charged particle moves in the region at constant velocity.

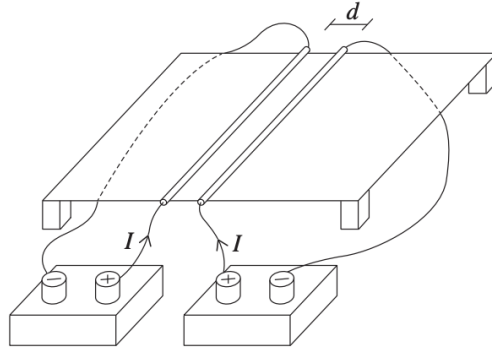


What is the direction of the particle's velocity?

- A. Up the page
- B. Down the page
- C. To the left
- D. To the right

3.2.10 Question 9 (HSC 2022 Q15)

- 15 Two wires separated by a distance, d , carry equal electric currents producing a magnetic force between them.



The separation between the wires is increased to $4d$ and the current in each wire is doubled.

What happens to the magnetic force between the wires, compared to the original force?

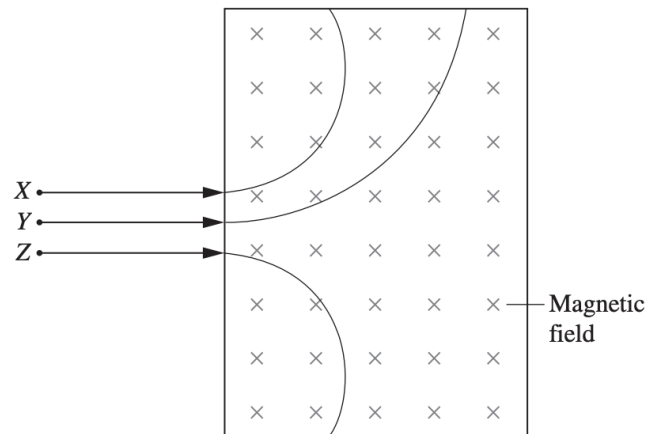
- A. It does not change.
- B. It increases by a factor of 4.
- C. It decreases by a factor of 4.
- D. It decreases by a factor of 8.

3.2.11 Question 10 (HSC 2022 Q34)

Question 34 (7 marks)

Three charged particles, X, Y and Z, travelling along straight, parallel trajectories at the same speed, enter a region in which there is a uniform magnetic field which causes them to follow the paths shown. Assume that the particles do not exert any significant force on each other.

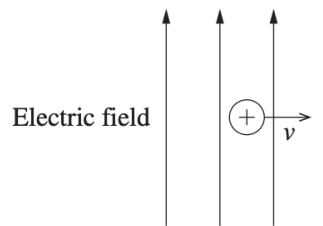
7



Explain the different paths that the particles follow through the magnetic field.

3.2.12 Question 11 (HSC 2021 Q2)

2 A positively charged particle is moving at velocity, v , in an electric field as shown.

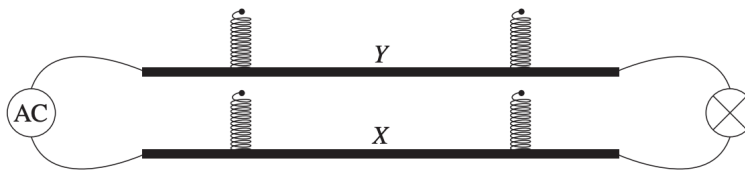


What is the direction of the force acting on the particle due to the electric field?

- A. Into the page
- B. Out of the page
- C. Up the page
- D. Down the page

3.2.13 Question 12 (HSC 2021 Q17)

- 17 Two long, parallel conductors X and Y are connected to a light bulb and an AC power supply. The conductors are suspended horizontally from fixed points using sensitive spring balances. X is positioned directly below Y .



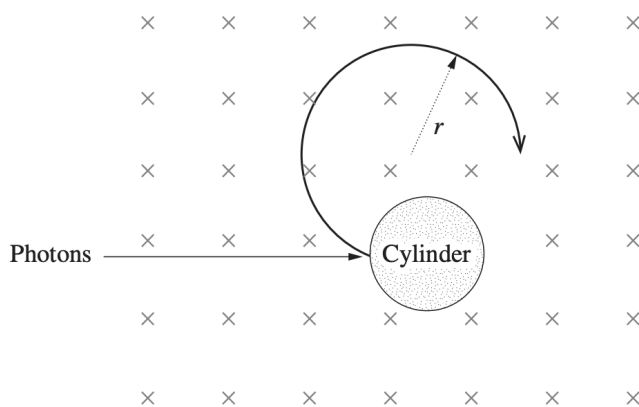
Which statement correctly compares the forces measured by the spring balances?

- A. The forces measured on X and Y will always be equal.
- B. The force measured on Y will be greater than or equal to that on X .
- C. The force measured on X will be greater than or equal to that on Y .
- D. There will be a continuous reversal of which measured force is greater.

3.2.14 Question 13 (HSC 2021 Q20)

- 20 A metal cylinder is located in a uniform magnetic field. The work function of the metal is ϕ .

Photons having an energy of 2ϕ strike the side of the cylinder, liberating photoelectrons which travel perpendicular to the magnetic field in a circular path. The maximum radius of the path is r .



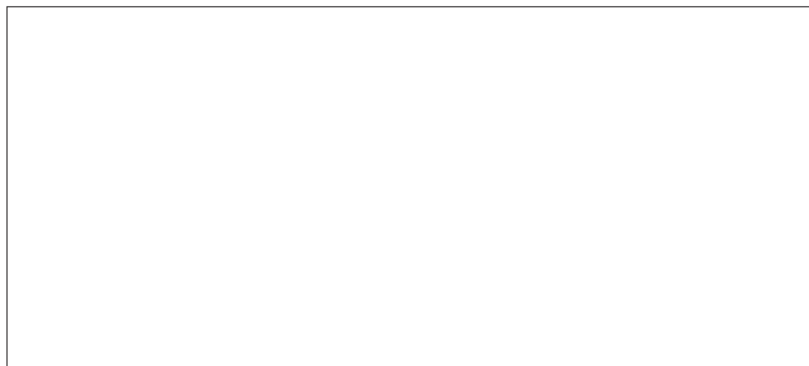
If the photon energy is doubled, what will the maximum radius of the path become?

- A. $2r$
- B. $3r$
- C. $\sqrt{2}r$
- D. $\sqrt{3}r$

3.2.15 Question 14 (HSC 2021 Q1)

A student is considering how to levitate a thin metal rod in a strong magnetic field of 1.2 T. The current flowing through the rod will be 2.3 A.

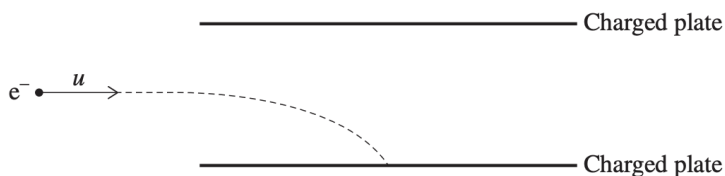
- (a) Use a labelled diagram to show a suitable orientation of the current and the magnetic field to achieve this outcome. Include relevant forces in your diagram. 3



- (b) Explain why the maximum mass per unit length of the rod cannot exceed 0.282 kg m^{-1} . Support your answer with a calculation. 3

3.2.16 Question 15 (HSC 2020 Q10)

- 10 An electron travelling in a straight line with an initial velocity, u , enters a region between two charged plates in which there is an electric field causing it to travel along the path as shown.



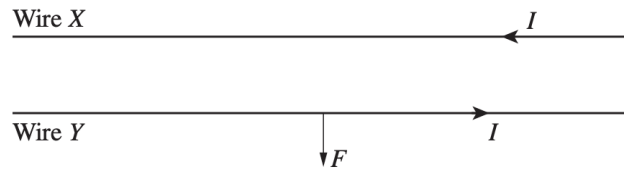
A magnetic field is then applied causing a second electron with the same initial velocity to pass through undeflected.

Which row of the table shows the directions of the electric and magnetic fields when the second electron enters the region between the plates?

	<i>Electric field</i>	<i>Magnetic field</i>
A.	Towards top of page	Into page
B.	Towards top of page	Out of page
C.	Towards bottom of page	Into page
D.	Towards bottom of page	Out of page

3.2.17 Question 16 (HSC 2020 Q1)

- 14 Two parallel wires, X and Y , each carry a current I .



The magnitude and direction of the force on wire Y are represented by the vector F .

The current in wire Y is then doubled and its direction is reversed. The current in wire X remains unchanged.

Which vector arrow represents the force on wire X after the change to the current in wire Y ?

A.



B.



C.



D.



3.2.18 Question 17 (HSC 2020 Q34)

Question 34 (6 marks)

A charged particle, q_1 , is fired midway between oppositely charged plates X and Y , as shown in Figure 1. The voltage between the plates is V volts.

The particle strikes plate Y at point P , a horizontal distance s from the edge of the plate. Ignore the effect of gravity.

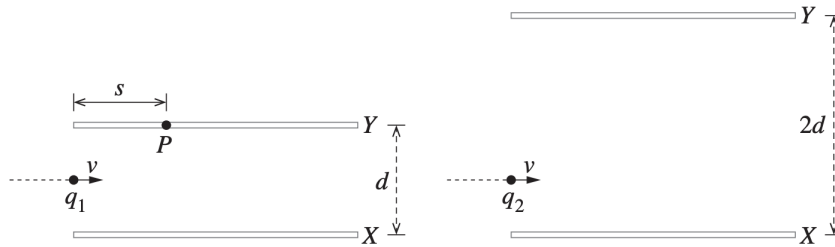


Figure 1

Figure 2

Plate Y is then moved to the position shown in Figure 2, with the voltage between the plates remaining the same.

An identical particle, q_2 , is fired into the electric field at the same velocity, entering the field at the same distance from plate X as q_1 .

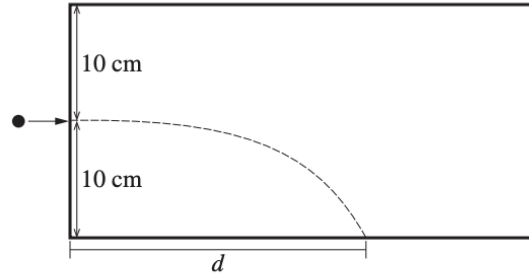
- (a) Compare the work done on q_1 and q_2 . 3

Question 34 (continued)

- (b) Compare the horizontal distances travelled by q_1 and q_2 in the electric field. 3

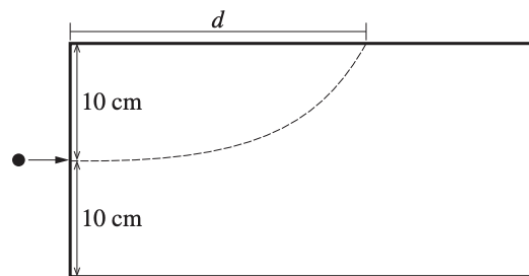
3.2.19 Question 18 (HSC 2019 Q16)

- 16 The diagram shows the trajectory of a particle with charge q and mass m when fired horizontally into a vacuum chamber, where it falls under the influence of gravity.



The horizontal distance, d , travelled by the particle is recorded.

The experiment is repeated with a uniform vertical electric field applied such that the particle travels the same horizontal distance, d , but strikes the upper surface of the chamber.

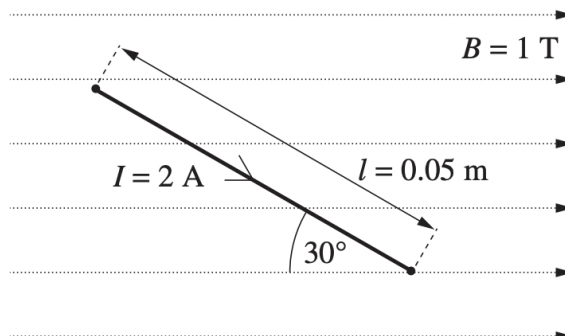


What is the magnitude of the electric field?

- A. mgq
- B. $2mgq$
- C. $\frac{mg}{q}$
- D. $\frac{2mg}{q}$

3.2.20 Question 19 (HSC 2018 Q5)

- 5 The diagram shows a current-carrying conductor in a magnetic field.

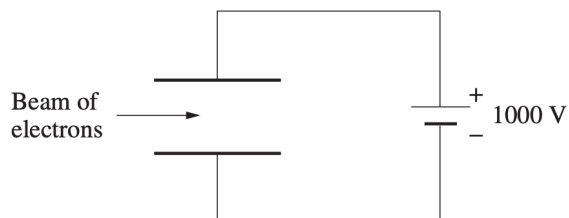


What is the magnitude of the force on the conductor?

- A. 0 N
- B. 0.05 N
- C. 0.09 N
- D. 0.10 N

3.2.21 Question 20 (HSC 2018 Q12)

- 12 The diagram shows electrons travelling in a vacuum at $2 \times 10^6 \text{ m s}^{-1}$ between two charged metal plates $1 \times 10^{-3} \text{ m}$ apart.



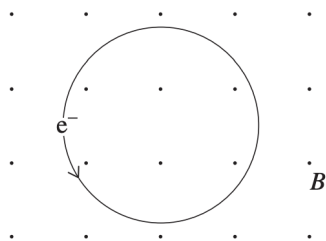
A magnetic field is to be applied to make the electrons continue to travel in a straight line.

What is the magnitude and direction of the magnetic field that is to be applied?

- A. $5 \times 10^{-1} \text{ T}$ into the page
- B. $5 \times 10^{-1} \text{ T}$ out of the page
- C. $1 \times 10^6 \text{ T}$ into the page
- D. $1 \times 10^6 \text{ T}$ out of the page

3.2.22 Question 21 (HSC 2018 Q13)

- 13 An electron moves in a circular path with radius r in a magnetic field as shown.



If the speed of the electron is increased, which row of the table correctly shows the effects of this change?

	<i>Force on electron</i>	<i>Radius of path</i>
A.	Increases	Decreases
B.	Increases	Increases
C.	Decreases	Decreases
D.	Decreases	Increases

3.2.23 Question 22 (HSC 2018 Q24)

Question 24 (5 marks)

- (a) Outline TWO features of high-voltage transmission lines that contribute to the safe transmission of electricity. 2

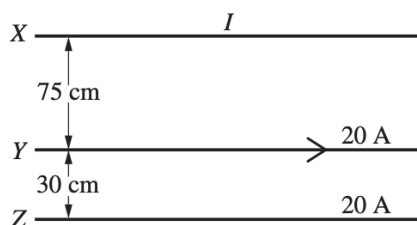
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- (b) Three parallel wires X , Y and Z all carry electric currents. A force of attraction is produced between Y and Z . There is zero net force on Y . 3



What is the magnitude and direction of the current in X ?

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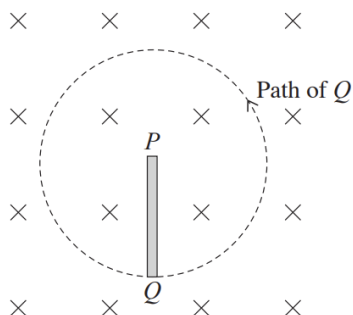
3.3 Faraday's Law

3.3.1 Question 1 (Textbook)

State Faraday's law, including the equation, and distinguish it from Lenz's law. What other physical concepts/laws is Lenz's law related to?

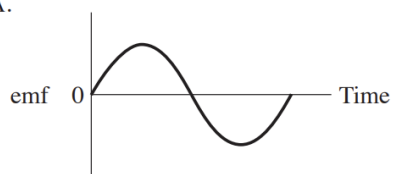
3.3.2 Question 2 (HSC 2024 Q15)

- 15 A uniform magnetic field is directed into the page. A conductor PQ rotates about the end P at a constant rate.

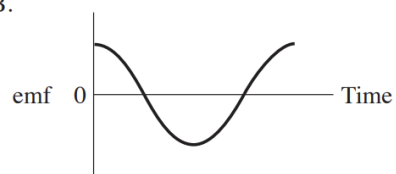


Which graph shows the emf induced between the ends of the conductor, P and Q , as it rotates one revolution from the position shown?

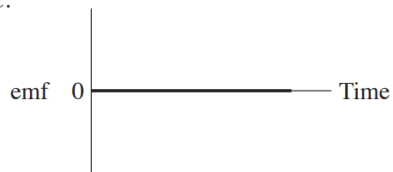
A.



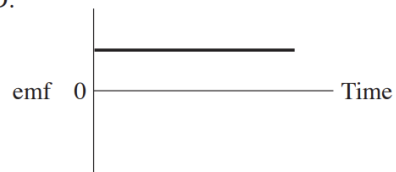
B.



C.

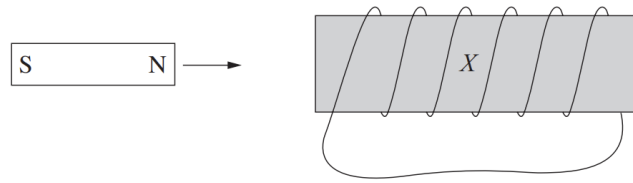


D.



3.3.3 Question 3 (HSC 2024 Q18)

- 18** The diagram shows a magnet moving towards a coil X .



This action causes a current to be induced in the coil.

Which situation will induce a current in coil X that is in the same direction as the current induced by the movement of the magnet?

A.

B.

C.

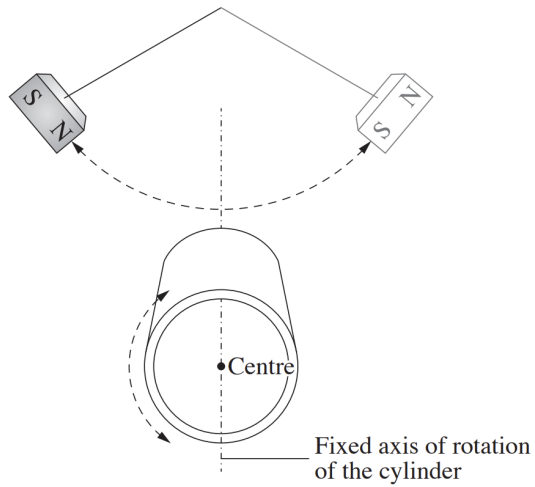
D.

3.3.4 Question 4 (HSC 2024 Q6)

Question 33 (7 marks)

A magnet is swinging as a pendulum. Close below it is an aluminium (non-ferromagnetic) can. The can is free to spin around a fixed axis as shown.

7

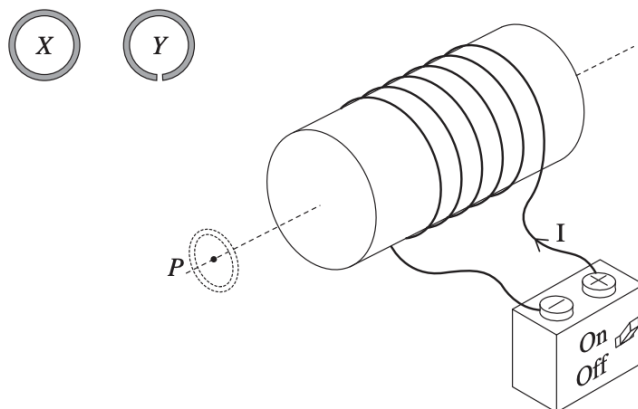


Analyse the motion and energy transformations of both the can and the magnet.

3.3.5 Question 5 (HSC 2023 Q30)

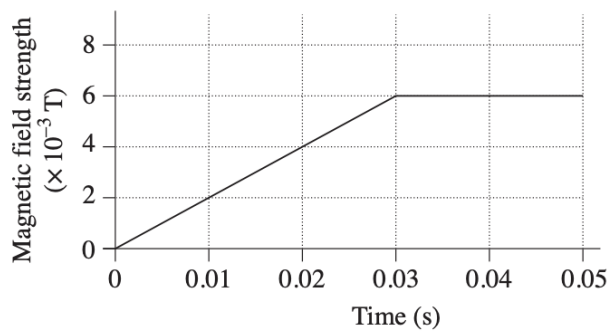
Question 30 (8 marks)

The diagram shows apparatus that is used to investigate the interaction between the magnetic field produced by a coil and two copper rings *X* and *Y*, when each is placed at position *P*, as shown.



Ring *X* is a complete circular ring, and a small gap has been cut in ring *Y*. Each of the rings has a cross-sectional area of $4 \times 10^{-4} \text{ m}^2$.

The power supply connected to the coil produces an increasing current through the coil in the direction shown, when the switch is turned on. This produces a magnetic field at *P* that varies as shown in the graph.



Question 30 (continued)

- (a) In the first part of the investigation, ring X is held near the end of the electromagnet at position P . 4

Account for the force acting on the ring from 0 to 0.05 seconds after the power supply is turned on.

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- (b) (i) In the second part of the investigation, ring Y is placed at P , and the power supply is turned on. 2

Explain the behaviour of the ring.

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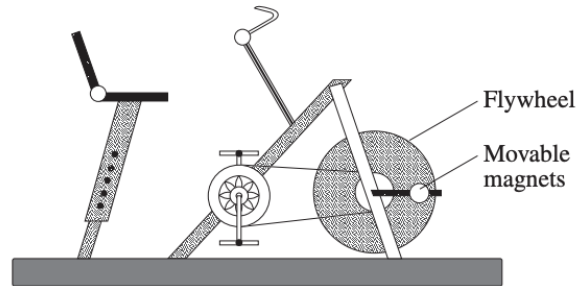
.....

- (ii) Calculate the maximum induced emf in ring Y . 2

3.3.6 Question 6 (HSC 2022 Q32)

Question 32 (6 marks)

One type of stationary exercise bike uses a pair of strong, movable magnets placed on opposite sides of a thick, aluminium flywheel to provide a torque to make it harder to pedal.



- (a) Explain the principle by which these magnets make it harder to pedal.

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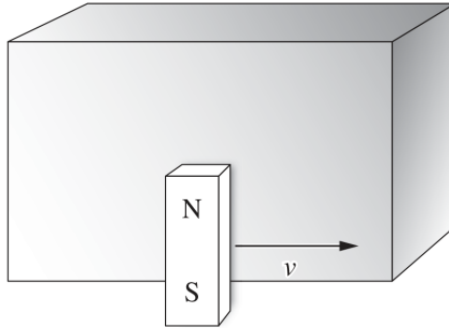
.....

- (b) The bike rider wants to increase the opposing torque on the flywheel. Justify an adjustment that could be made to the magnets to achieve this.

3

3.3.7 Question 7 (HSC 2021 Q10)

- 10 A strong magnet is moved past a copper block at a constant speed as shown.



What is the direction of the force acting on the copper block?

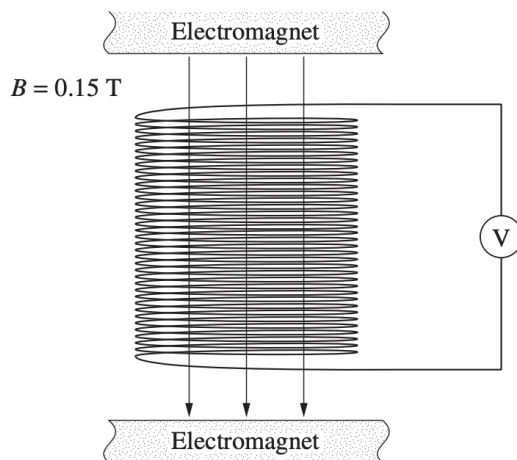
- A. To the left
- B. To the right
- C. Into the page
- D. Out of the page

3.3.8 Question 8 (HSC 2021 Q24)

Question 24 (3 marks)

A stationary coil of 35 turns and cross-sectional area of 0.02 m^2 is placed between two electromagnets, and connected to a voltmeter as shown. The electromagnets produce a uniform magnetic field of 0.15 T through the coil.

3



The magnitude of the magnetic field is then reduced to zero at a constant rate over a period of 0.4 s .

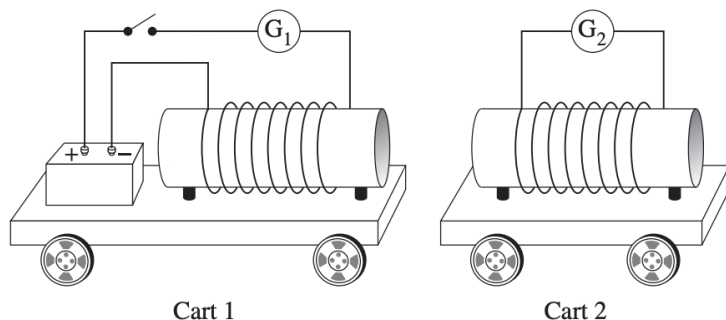
Calculate the magnitude of the emf induced in the coil.

3.3.9 Question 9 (HSC 2021 Q1)

Question 31 (7 marks)

Two identical solenoids are mounted on carts as shown. Each solenoid is connected to a galvanometer, and the solenoid on cart 1 is also connected to an open switch and a battery. The total mass of cart 1 is twice that of cart 2.

7



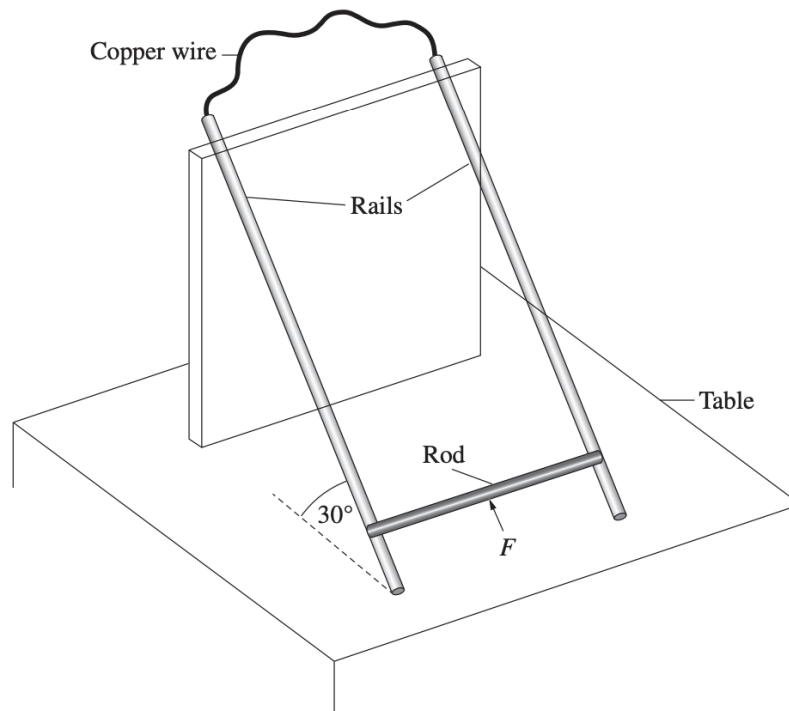
Explain what would be observed when the switch on cart 1 is closed. In your answer, refer to the current in each galvanometer and the initial movement of the carts.

3.3.10 Question 10 (HSC 2020 Q28)

Question 28 (7 marks)

A metal rod sits on a pair of parallel metal rails, 20 cm apart, that are connected by a copper wire. The rails are at 30° to the horizontal.

The apparatus is in a uniform magnetic field of 1 T which is upward, perpendicular to the table.



A force, F , is applied parallel to the rails to move the rod at a constant speed along the rails. The rod is moved a distance of 30 cm in 2.5 s.

- (a) Show that the change in magnetic flux through the circuit while the rod is moving is approximately 5.2×10^{-2} Wb.

2

Question 28 (continued)

- (b) Calculate the emf induced between the ends of the rod while it is moving, and state the direction of flow of the current in the circuit. **2**

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- (c) The experiment is repeated without the magnetic field. **3**

Explain why the force required to move the rod is different without the magnetic field.

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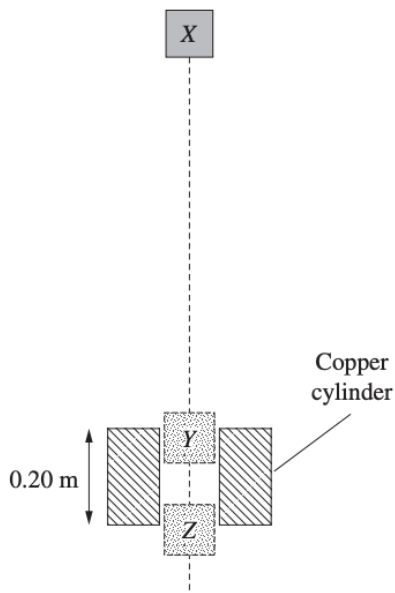
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3.3.11 Question 11 (HSC 2020 Q31)

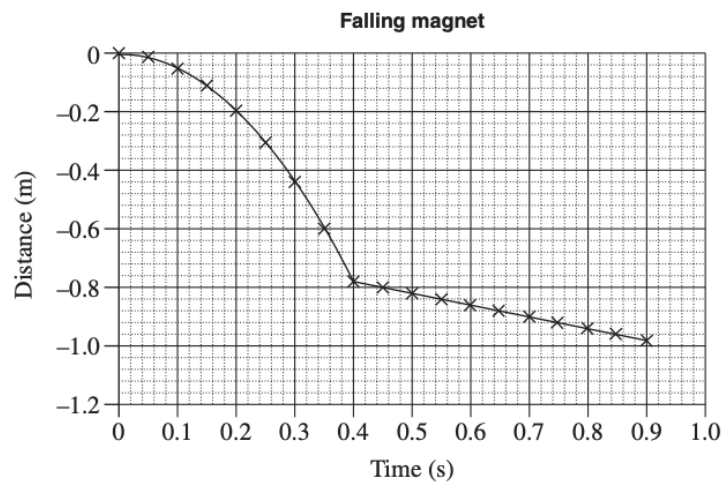
Question 33 (9 marks)

A strong magnet of mass 0.04 kg falls 0.78 m under the action of gravity from position X above a hollow copper cylinder. It then travels a distance of 0.20 m through the cylinder from Y to Z before falling freely again.

9



The magnet takes 0.5 seconds to pass through the cylinder. The displacement–time graph of the magnet is shown.



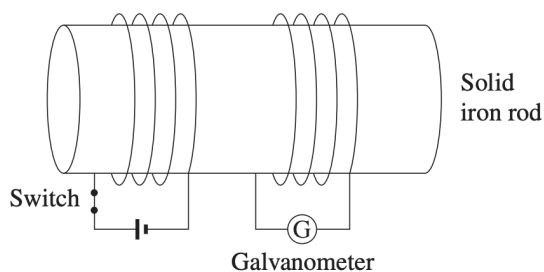
Question 33 (continued)

Analyse the motion of the magnet by applying the law of conservation of energy.

Your analysis should refer to gravity and the copper cylinder, and include both qualitative and quantitative information.

3.3.12 Question 12 (HSC 2019 Q5)

- 5 The diagram shows two coils wound around a solid iron rod. Initially the switch is closed.



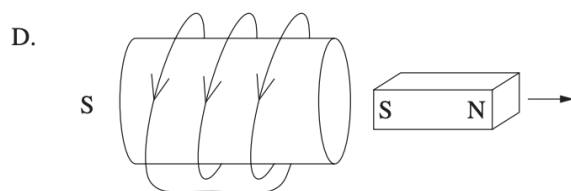
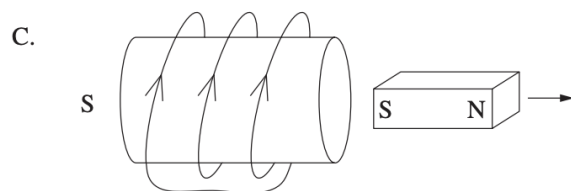
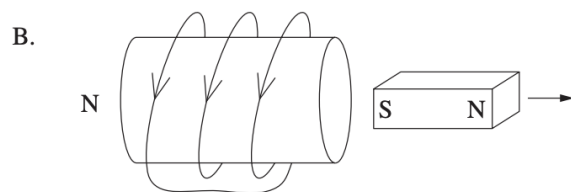
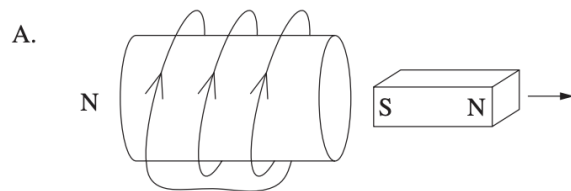
Opening the switch will cause the galvanometer pointer to

- A. remain at a constant reading.
- B. move from a non-zero reading to a zero reading.
- C. move from a zero reading to a non-zero reading, where it remains.
- D. move from a zero reading to a non-zero reading, then back to zero.

3.3.13 Question 13 (HSC 2019 Q7)

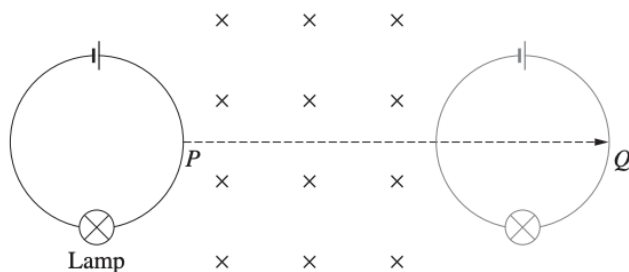
- 7 A bar magnet is moved away from a stationary coil.

Which diagram correctly shows the direction of the induced current in the coil and the resulting magnetic polarity of the coil?

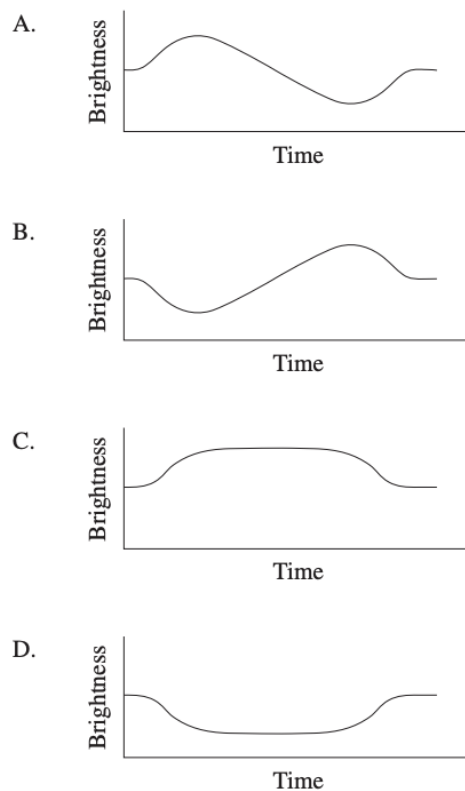


3.3.14 Question 14 (HSC 2019 Q18)

- 18 A circular loop of wire is connected to a battery and a lamp. The apparatus is moved from P to Q along the path shown at a constant velocity through a region containing a uniform magnetic field.



Which graph shows the brightness of the lamp as the apparatus moves between P and Q ?

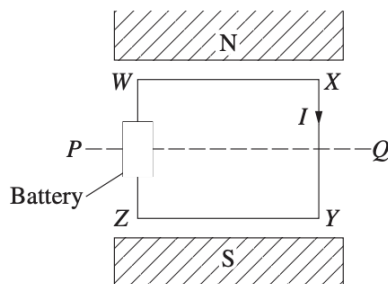


3.3.15 Question 15 (HSC 2019 Q28)

Question 28 (3 marks)

A metal loop, $WXYZ$ is connected to a battery and placed in a uniform magnetic field. A current flows through the loop in the direction shown.

3



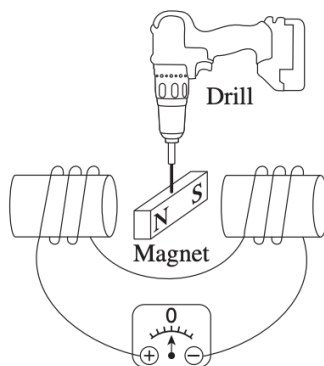
The loop is then allowed to rotate by 90° about the axis PQ .

Compare the forces acting on WX and XY before and after this rotation.

3.3.16 Question 16 (HSC 2018 Q22)

Question 22 (continued)

- (b) The diagram shows a magnet attached to an electric drill so that it can be rotated between two coils connected to a voltmeter. 3



The drill starts from rest and gradually speeds up, reaching its full speed after three revolutions.

Sketch a graph showing the induced emf across the coils during the time that it takes the magnet to reach its full speed.



3.4 Transformers

3.4.1 Question 1 (HSC 2024 Q8)

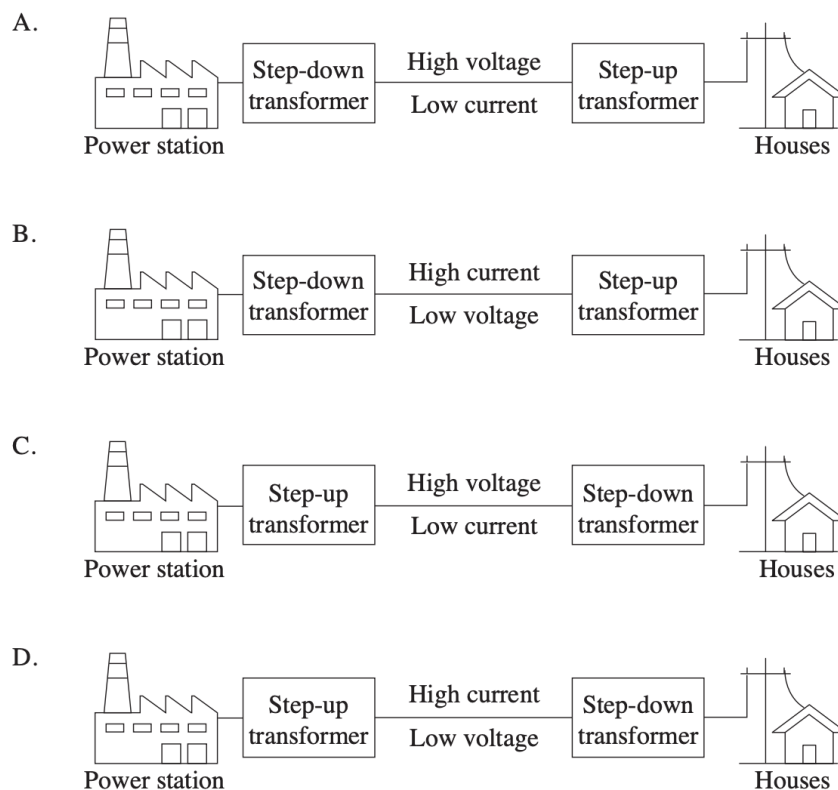
- 8 An ideal transformer produces an output of 6 volts when an input of 240 volts is applied.

What change would be needed to produce an output of 12 volts, using the same input voltage?

- A. Increase the number of turns on the primary coil
- B. Decrease the number of turns on the primary coil
- C. Increase the resistance connected to the secondary coil
- D. Decrease the resistance connected to the secondary coil

3.4.2 Question 2 (HSC 2023 Q2)

- 2 Which diagram best represents the transmission of energy from a power station to people's houses?

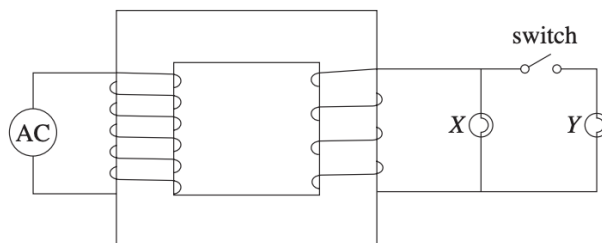


3.4.3 Question 3 (HSC 2023 Q28)

Question 28 (5 marks)

An ideal transformer is connected to a 240 V AC supply. It has 300 turns on the primary coil and 50 turns on the secondary coil.

It is connected in the circuit with two identical light globes, X and Y , as shown.



- (a) Calculate the voltage across light globe X when the switch is open. 2
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-
- (b) Explain why, after the switch has been closed, the current in the primary coil is different from when the switch is open. 3

3.4.4 Question 4 (HSC 2022 Q1)

- 1 An ideal transformer has 20 turns on the primary coil and an input voltage of 100 V.

How many turns are there on the secondary coil if the output voltage is 400 V?

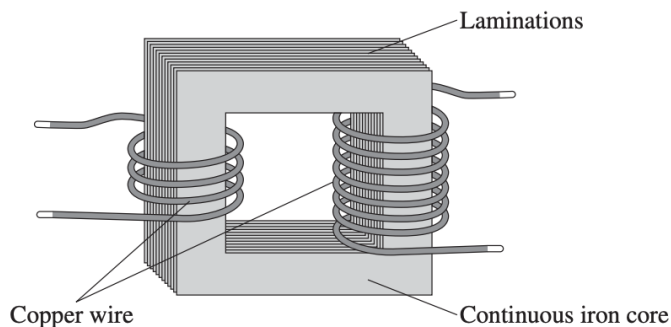
- A. 4
- B. 5
- C. 80
- D. 400

3.4.5 Question 5 (HSC 2022 Q22)

Question 22 (4 marks)

The diagram shows features of a transformer.

4



For TWO features of the transformer, describe how each contributes to the transformer's efficiency.

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3.4.6 Question 6 (HSC 2021 Q7)

- 7 In a certain ideal transformer, the current in the secondary coil is four times as large as the current in the primary coil.

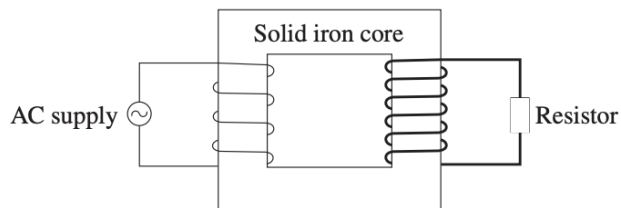
Which row of the table correctly identifies the type of transformer and the ratio of turns?

	Type of transformer	Ratio of turns in primary coil to turns in secondary coil
A.	Step up	4 : 1
B.	Step up	1 : 4
C.	Step down	4 : 1
D.	Step down	1 : 4

3.4.7 Question 7 (HSC 2019 Q24)

Question 24 (7 marks)

A step-up transformer is constructed using a solid iron core. The coils are made using copper wires of different thicknesses as shown.



The table shows electrical data for this transformer.

V_s	I_s	$V_p I_p$
50 V	9 A	500 J s^{-1}

- (a) Explain how the operation of this transformer remains consistent with the law of conservation of energy. Include a relevant calculation in your answer.

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- (b) Explain how TWO modifications to this transformer would improve its efficiency.

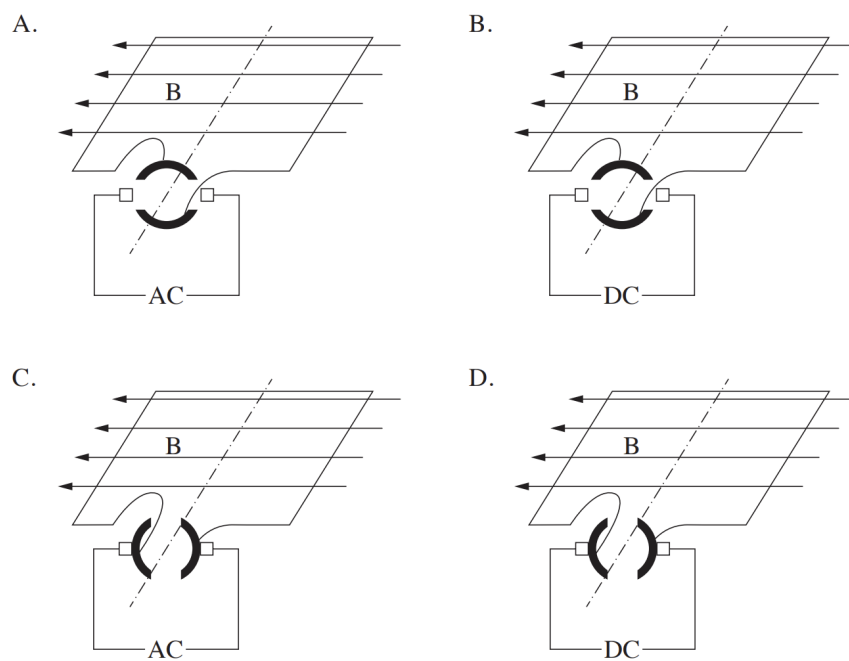
4

3.5 Motors

3.5.1 Question 1 (HSC 2024 Q4)

- 4 A conducting coil is mounted on an axle and placed in a uniform magnetic field. The diagram shows different ways of connecting the coil to a power source.

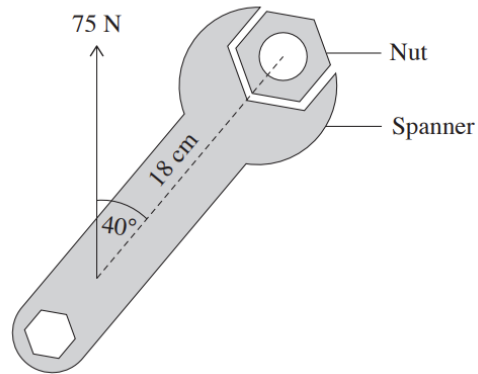
Which setup allows the conducting coil to rotate continuously?



3.5.2 Question 2 (HSC 2024 Q21)

Question 21 (6 marks)

To tighten a nut, a force of 75 N is applied to a spanner at an angle, as shown.



- (a) Calculate the magnitude of the torque produced by the applied force.

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- (b) Explain TWO ways in which torque can be increased in a simple DC motor.

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3.5.3 Question 3 (HSC 2023 Q10)

- 10 Figure I shows a current flowing through a loop of wire that is in a uniform magnetic field.

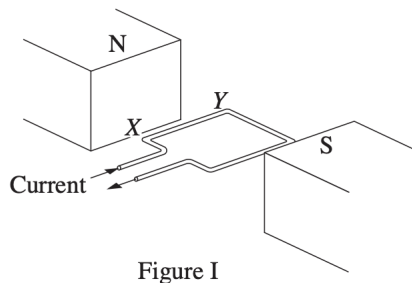


Figure I

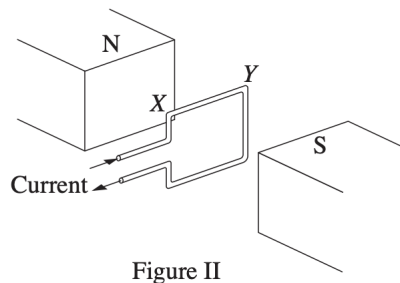


Figure II

The loop is then rotated to the position shown in Figure II.

The magnitude of the force on the side XY and the magnitude of the torque on the loop in Figure II are compared to those in Figure I.

Which row of the table correctly describes the comparison?

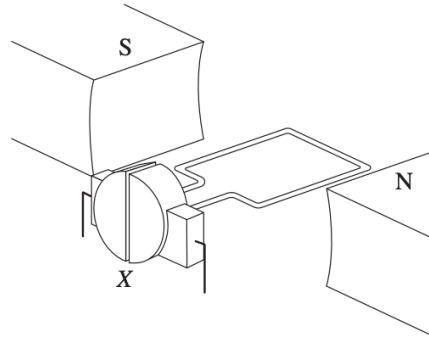
	<i>Force</i>	<i>Torque</i>
A.	$I > II$	$I = II$
B.	$I > II$	$I > II$
C.	$I = II$	$I = II$
D.	$I = II$	$I > II$

3.5.4 Question 4 (HSC 2023 Q25)

Question 25 (4 marks)

- (a) The diagram represents one type of electric motor.

2



Describe the function of part X.

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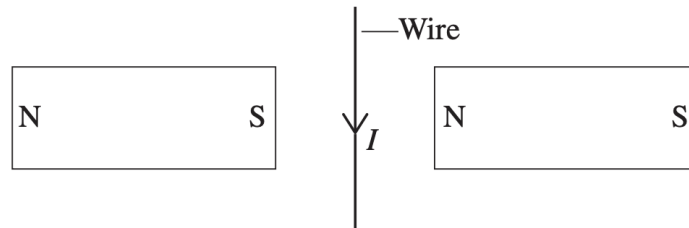
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- (b) Explain why the torque of a DC motor decreases as its rotational speed increases.

2

3.5.5 Question 5 (HSC 2022 Q19)

- 4 A current-carrying wire is in a magnetic field, as shown.

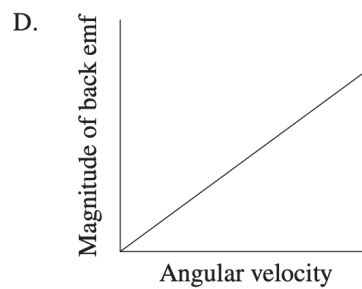
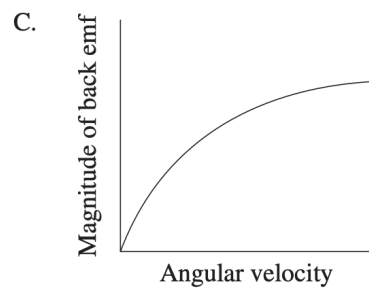
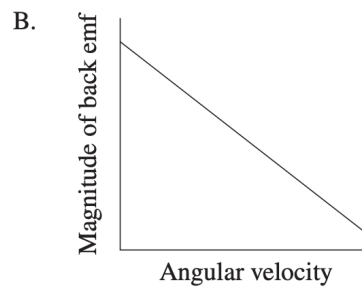
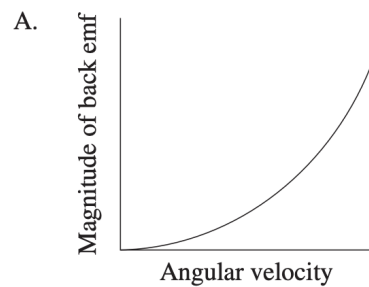


What is the direction of the force on the wire?

- A. Left
- B. Right
- C. Into the page
- D. Out of the page

3.5.6 Question 6 (HSC 2021 Q12)

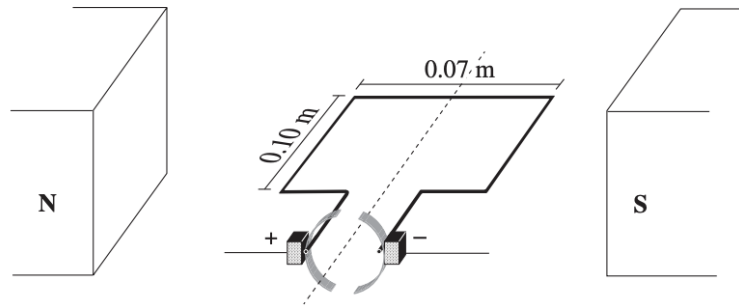
- 12 Which graph shows the magnitude of back emf induced in a DC motor rotating continuously at different angular velocities?



3.5.7 Question 7 (HSC 2021 Q21)

Question 21 (4 marks)

A DC motor is constructed from a single loop of wire with dimensions $0.10\text{ m} \times 0.07\text{ m}$. The magnetic field strength is 0.40 T and a current of 14 A flows through the loop.



- (a) Calculate the magnitude of the maximum torque produced by the motor. 2

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- (b) Describe how the magnitude of the torque changes as the loop moves through half a rotation from the position shown. 2

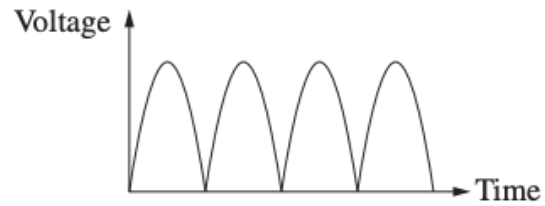
3.5.8 Question 8 (HSC 2020 Q2)

- 2 Which of the following is NOT required for the operation of AC induction motors?

- A. Brushes
- B. Stator winding
- C. Magnetic fields
- D. Current applied to the rotor

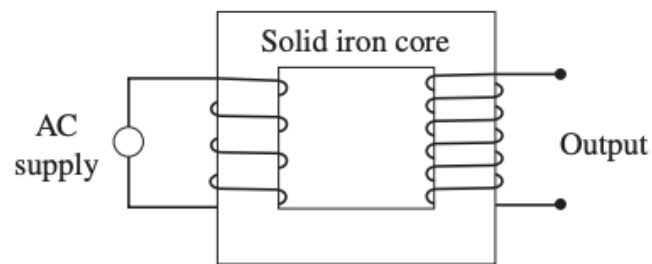
3.5.9 Question 9 (HSC 2020 Q7)

7 The output of a device is shown.

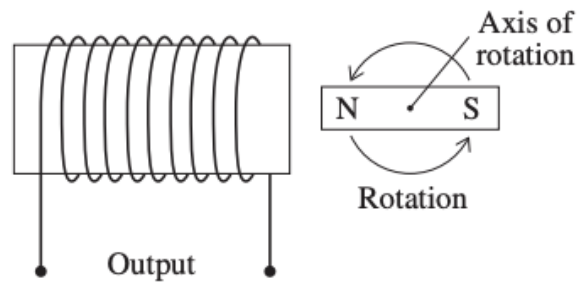


Which diagram represents the device that has the output shown?

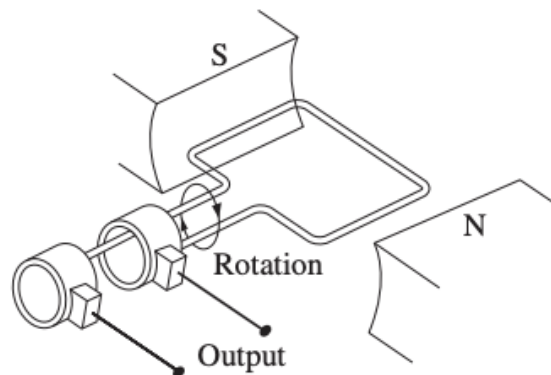
A.



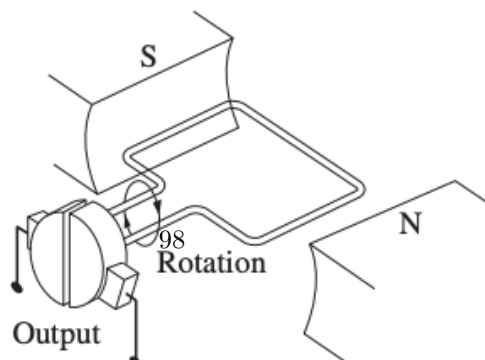
B.



C.



D.

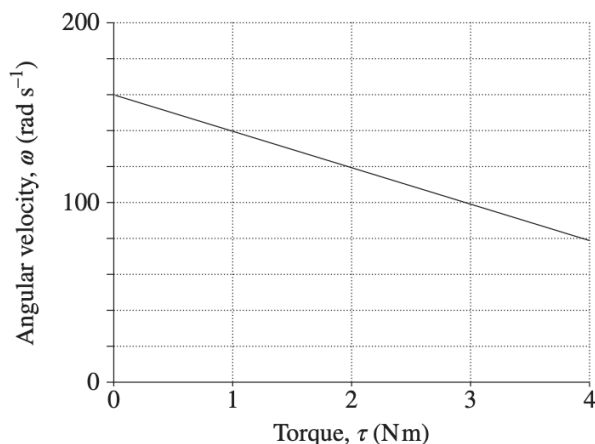


3.5.10 Question 10 (HSC 2020 Q23)

Question 23 (3 marks)

The graph shows data for a motor connected to a 240 V power supply.

3



The equation for the torque, τ , produced by the motor is $\tau = \frac{VI\eta}{\omega}$

where τ = torque (Nm)
 V = voltage (V)
 I = current (A)
 η = efficiency = 0.3
 ω = angular velocity (rad s⁻¹)

A circuit breaker cuts the current to the motor if the current exceeds 5 A.

Determine what will happen when the motor produces a torque of 2.95 Nm. Show relevant calculations.

3.5.11 Question 11 (HSC 2018 Q4)

- 4 A motor, battery and ammeter are connected in series. When the motor is turning at full speed, the ammeter has a reading of 0.1 A. While the motor is spinning, a person holds the shaft of the motor to stop it.

Which row of the table correctly identifies the change in the ammeter reading and an explanation for the change?

	Reading on ammeter	Explanation
A.	Decreases	Decrease in back emf
B.	Increases	Increase in back emf
C.	Decreases	Increase in back emf
D.	Increases	Decrease in back emf

4 Light

4.1 History of light

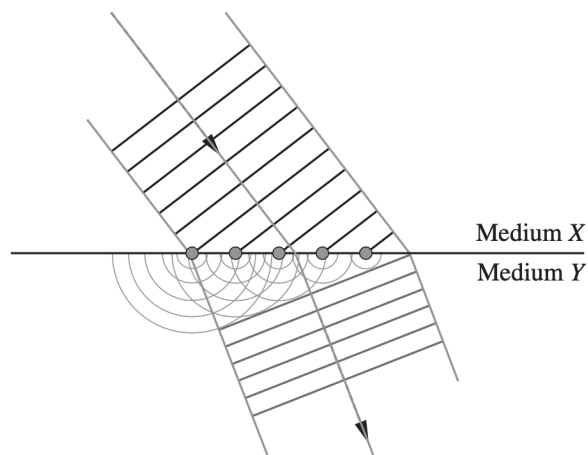
4.1.1 Question 1 (HSC 2024 Q2)

2 Which of the following provides evidence for the model of light proposed by Huygens?

- A. Emission spectra
- B. Diffraction of light
- C. Black body radiation
- D. The photoelectric effect

4.1.2 Question 2 (HSC 2020 Q1)

1 The diagram shows a model used to explain the refraction of light passing from medium X into medium Y.



Who proposed this model?

- A. Malus
- B. Planck
- C. Newton
- D. Huygens

4.1.3 Question 3 (HSC 2020 Q3)

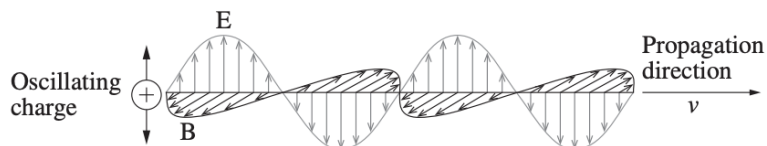
- 3 What was the basis for Maxwell's prediction of the velocity of electromagnetic waves?
- A. Experiments using magnetic fields to accelerate particles
 - B. Experiments using light and mirrors to establish the finite speed of light
 - C. Equations showing how oscillating electric and magnetic fields propagate
 - D. Equations showing how electromagnetic waves are affected by gravitational fields

4.1.4 Question 4 (HSC 2019 Q25)

Question 25 (4 marks)

The diagram shows a model of electromagnetic waves.

4



Relate this model to predictions made by Maxwell.

4.2 Polarisation of light

4.2.1 Question 1 (HSC 2023 Q29)

Question 29 (4 marks)

When light from an incandescent lamp is passed through a plane polarising filter, the intensity of the light is reduced.

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Explain this phenomenon.

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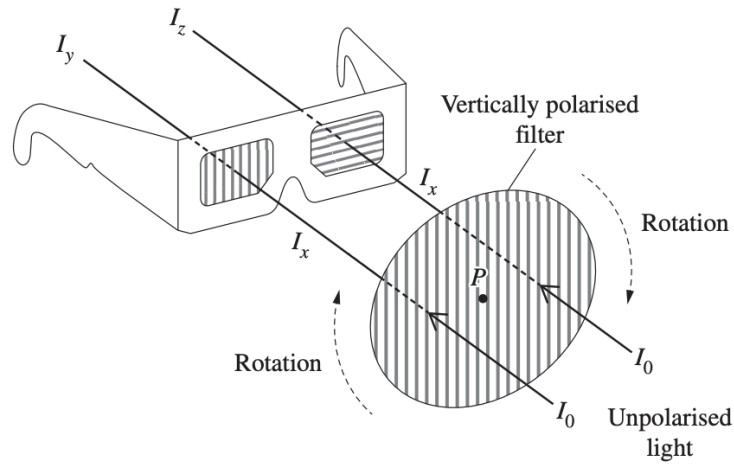
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4.2.2 Question 2 (HSC 2022 Q17)

- 17 Unpolarised light of intensity I_0 is incident upon a vertically polarised filter. The filtered light then passes through a pair of glasses. The glasses have polarising filters, with one side polarised vertically and the other horizontally.

The filter undergoes one complete 360° rotation around point P , as shown.

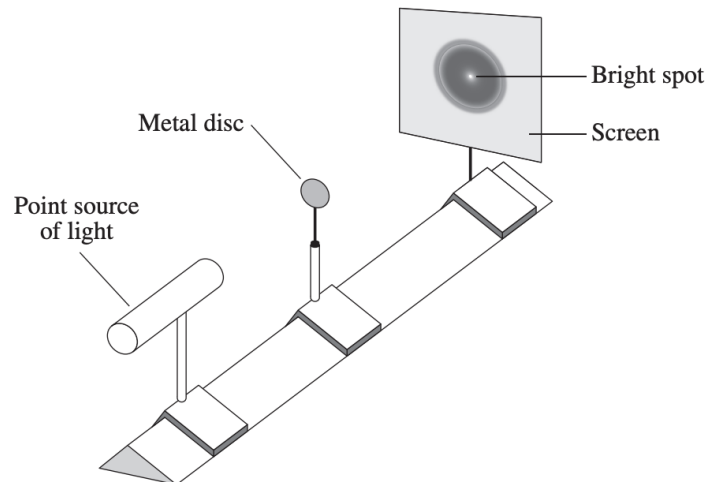


Which of the following correctly compares I_y to the intensity at other positions?

- A. I_y never equals I_x
- B. I_y never equals I_z
- C. I_y sometimes equals I_z
- D. I_y sometimes equals I_0

4.2.3 Question 3 (HSC 2021 Q8)

- 8 Light from a point source is incident upon a circular metal disc, forming a shadow on a screen as shown. A bright spot is observed in the centre of the shadow.

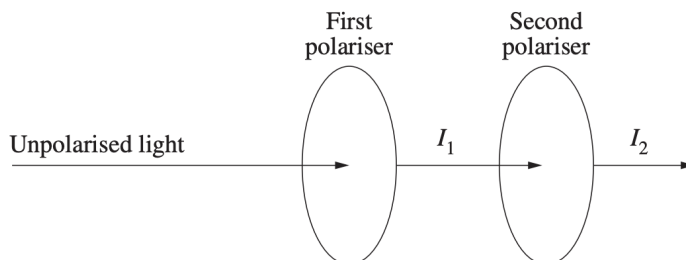


The bright spot is caused by a combination of

- A. interference and refraction.
- B. refraction and polarisation.
- C. polarisation and diffraction.
- D. diffraction and interference.

4.2.4 Question 4 (HSC 2021 Q15)

- 15 Unpolarised light is incident upon two consecutive polarisers as shown. The second polariser has a fixed transmission axis which cannot be rotated. I_1 is the intensity of light after the first polariser, and I_2 is the intensity of light after the second polariser.

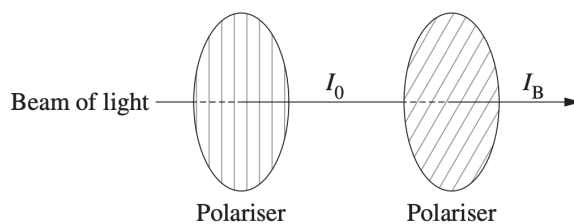


How would I_1 and I_2 be affected if the transmission axis of the first polariser was rotated?

- A. Both would change.
- B. Only I_1 would change.
- C. Only I_2 would change.
- D. Neither would change.

4.2.5 Question 5 (HSC 2019 Q10)

- 10 A beam of light passes through two polarisers. The second polariser has a transmission axis at an angle of 30° to that of the first polariser. The intensity of the light beam before and after the second polariser is I_0 and I_B respectively.



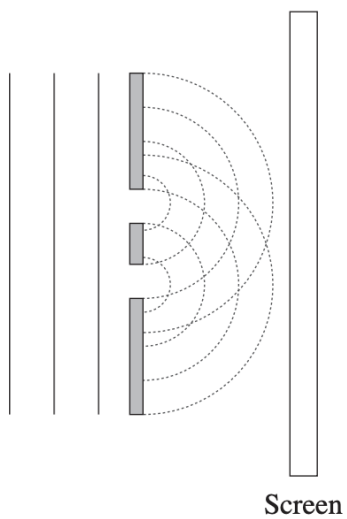
Which row of the table correctly identifies the value of $\frac{I_B}{I_0}$, and the model of light demonstrated by this investigation?

	Value of $\frac{I_B}{I_0}$	Model of light demonstrated
A.	0.750	Wave model
B.	0.750	Particle model
C.	0.866	Wave model
D.	0.866	Particle model

4.3 Double Slit Experiment

4.3.1 Question 1 (HSC 2023 Q3)

- 3 A diagram representing a double slit experiment using light is shown.



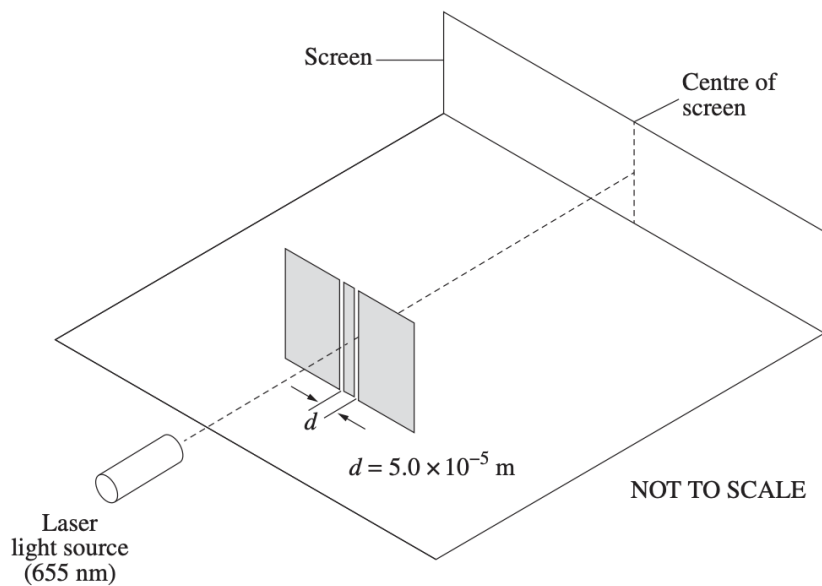
Which of the following best represents the expected pattern on the screen?

- A. 
- B. 
- C. 
- D. 

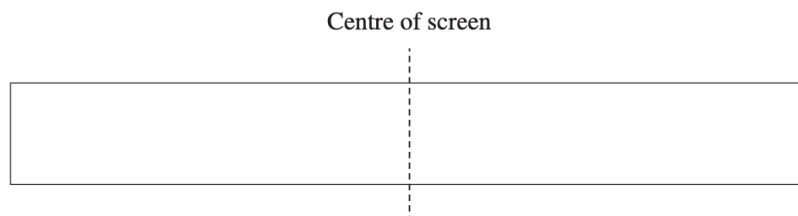
4.3.2 Question 2 (HSC 2022 Q27)

Question 27 (7 marks)

A laser producing red light of wavelength 655 nm is directed onto double slits separated by a distance, $d = 5.0 \times 10^{-5}$ m. A screen is placed behind the double slits.

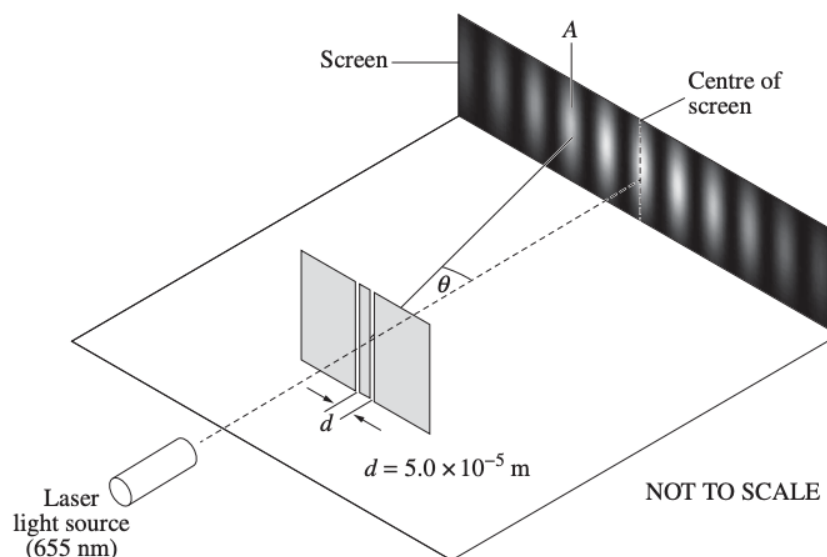


- (a) Newton proposed a model of light. Use a labelled sketch to show the pattern on the screen that would be expected from Newton's proposed model. 2



Question 27 (continued)

When the laser light is turned on, a series of vertical bright lines are seen on the screen.



- (b) Calculate the angle, θ , between the centre line and the bright line at A.

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- (c) The laser is replaced with one producing green light of wavelength 520 nm.

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Explain the difference in the pattern that would be produced.

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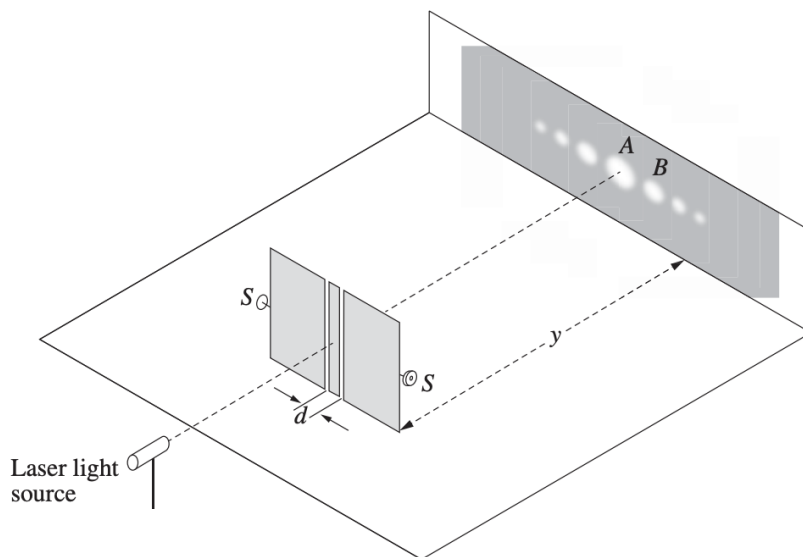
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4.3.3 Question 3 (HSC 2020 Q27)

Question 27 (4 marks)

The following apparatus is used to investigate light interference using a double slit.

4

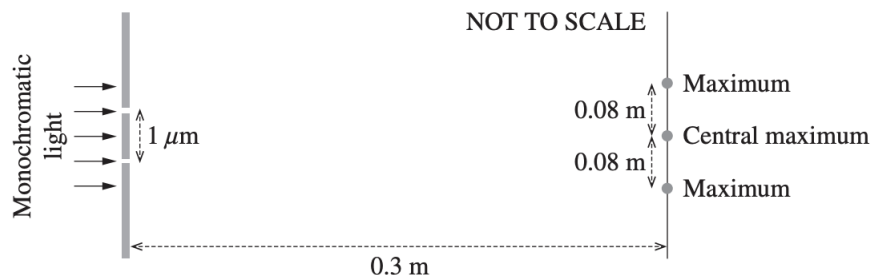


The distance, y , from the slits to the screen can be varied. The adjustment screws (S) vary the distance, d , between the slits. The wavelength of the laser light can be varied across the visible spectrum. The diffraction pattern shown is for a specific wavelength of light.

Explain TWO methods of keeping the distance between the maxima at A and B constant when the wavelength of the laser light is reduced.

4.3.4 Question 4 (HSC 2019 Q15)

- 15 Monochromatic light passes through two slits $1\ \mu\text{m}$ apart. The resulting diffraction pattern is measured at a distance of $0.3\ \text{m}$.



This diffraction pattern can be analysed using the equation $d \sin \theta = \lambda$.

What values of d and θ should be used in the equation?

	d	θ
A.	$0.3\ \text{m}$	$\tan^{-1}\left(\frac{0.08}{0.3}\right)$
B.	$0.3\ \text{m}$	$\sin^{-1}\left(\frac{0.08}{0.3}\right)$
C.	$1\ \mu\text{m}$	$\tan^{-1}\left(\frac{0.08}{0.3}\right)$
D.	$1\ \mu\text{m}$	$\sin^{-1}\left(\frac{0.08}{0.3}\right)$

4.4 Photoelectric effect

4.4.1 Question 1 (HSC 2024 Q6)

- 6 The photoelectric effect is mathematically modelled by the following relationship:

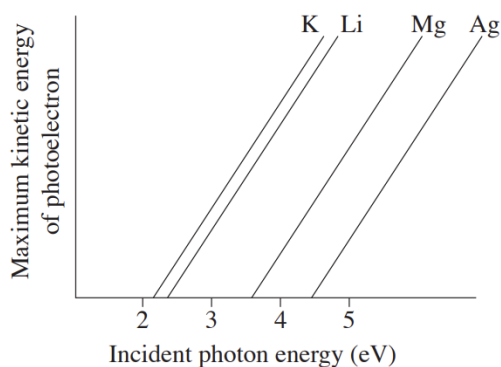
$$K_{\text{max}} = hf - \phi$$

In this model, the symbol ϕ represents the amount of energy

- A. supplied by a photon to an electron.
- B. retained by an electron after being hit.
- C. required to release an electron from a material.
- D. left over after a collision of a photon with an electron.

4.4.2 Question 2 (HSC 2024 Q16)

- 16 The graph shows the relationship between the maximum kinetic energy of emitted photoelectrons and the incident photon energy for four different metal surfaces.



Light of frequency 7×10^{14} Hz is incident on the metals.

From which metals are photoelectrons emitted?

- A. K, Li only
- B. Mg, Ag only
- C. All of the metals
- D. None of the metals

4.4.3 Question 3 (HSC 2022 Q7)

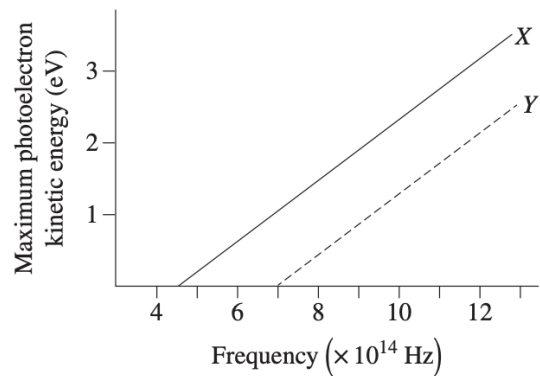
- 7 A photon has an energy of 9.0×10^{-24} J.

What is the frequency of this radiation?

- A. 1.00×10^{-40} Hz
- B. 7.36×10^{-11} Hz
- C. 1.36×10^{10} Hz
- D. 5.97×10^{11} Hz

4.4.4 Question 4 (HSC 2022 Q14)

- 14 Line X shows the results of an experiment carried out to investigate the photoelectric effect.



What change to this experiment would produce the results shown by line Y?

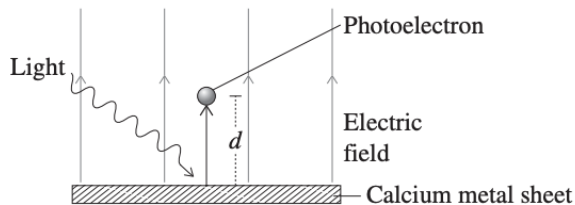
- A. Increasing the frequency of the radiation
- B. Using a metal that has a greater work function
- C. Decreasing the intensity of the incident radiation
- D. Decreasing the maximum energy of photoelectrons

4.4.5 Question 5 (HSC 2022 Q26)

Question 26 (6 marks)

Light of frequency 7.5×10^{14} Hz is incident on a calcium metal sheet which has a work function of 2.9 eV. Photoelectrons are emitted.

The metal is in a uniform electric field of 5.2 NC^{-1} , perpendicular to the surface of the metal, as shown.



- (a) Show that the maximum kinetic energy of an emitted photoelectron is $3.2 \times 10^{-20} \text{ J}$. 3

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- (b) Calculate the maximum distance, d , an emitted photoelectron can travel from the surface of the metal. 3

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4.4.6 Question 6 (HSC 2021 Q26)

Question 26 (6 marks)

A student performs an experiment to measure Planck's constant, h , using a device that emits specific frequencies of light when specific voltages are applied.

The voltage, V , needed to produce each frequency, f , is given by

$$V = \frac{hf}{q_e}$$

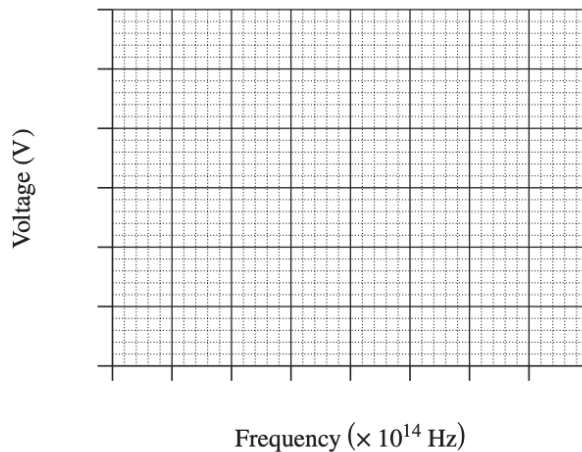
where q_e is the charge on an electron.

Data from the experiment is shown.

<i>Data point</i>	<i>Frequency</i> ($\times 10^{14}$ Hz)	<i>Voltage</i> (V)
1	3.5	1.3
2	4.8	1.7
3	5.3	1.9
4	7.0	2.6

- (a) Graph this data on the axes provided. Include a line of best fit.

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Question 26 (continued)

- (b) The student proposes using data point 1 to calculate a value for Planck's constant. **3**

Justify a better method to calculate Planck's constant from the experimental data provided.

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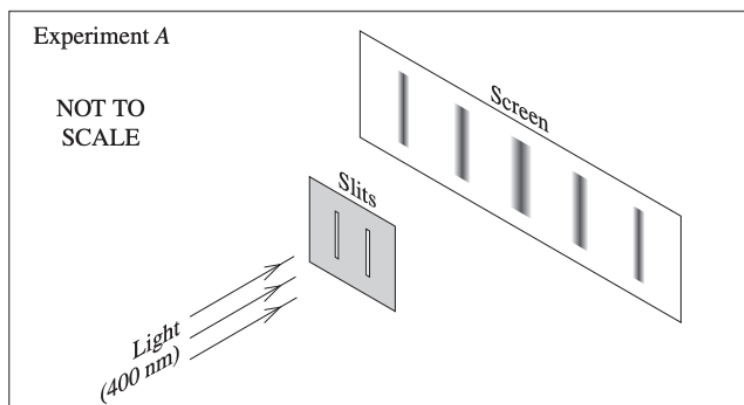
4.4.7 Question 7 (HSC 2021 Q33)

Question 33 (9 marks)

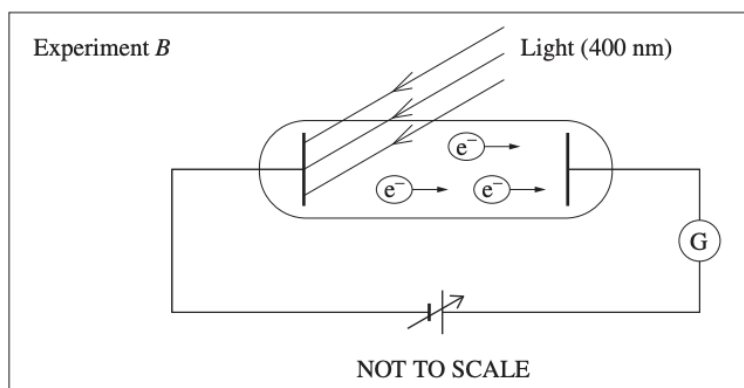
Two experiments are performed with identical light sources having a wavelength of 400 nm.

9

In experiment A, the light is incident on a pair of narrow slits 5.0×10^{-5} m apart, producing a pattern on a screen located 3.0 m behind the slits.



In experiment B, the light is incident on different metal samples inside an evacuated tube as shown. The kinetic energy of any emitted photoelectrons can be measured.



Some results from experiment B are shown.

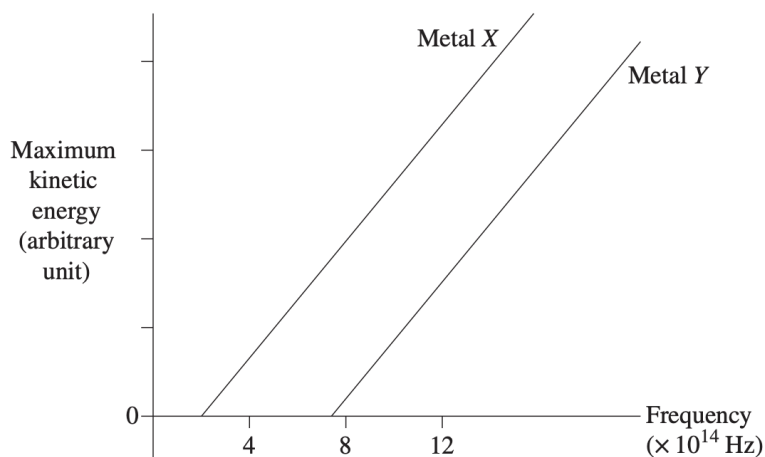
<i>Metal sample</i>	<i>Work function (J)</i>	<i>Photoelectrons observed?</i>
Nickel	8.25×10^{-19}	No
Calcium	4.60×10^{-19}	Yes

Question 33 (continued)

How do the results from Experiment *A* and Experiment *B* support TWO different models of light? In your answer, include a quantitative analysis of each experiment.

4.4.8 Question 8 (HSC 2020 Q13)

- 13 The graph shows the relationship between the frequency of light used to irradiate two different metals, and the maximum kinetic energy of photoelectrons emitted.



Suppose that light having a frequency of 8×10^{14} Hz is used to irradiate both metals.

Compared to the photoelectrons emitted from metal X, photoelectrons emitted from metal Y will

- A. have a lower maximum velocity.
- B. have a higher maximum velocity.
- C. take a longer time to gain sufficient energy to be ejected.
- D. take a shorter time to gain sufficient energy to be ejected.

4.4.9 Question 9 (HSC 2019 Q23)

Question 23 (3 marks)

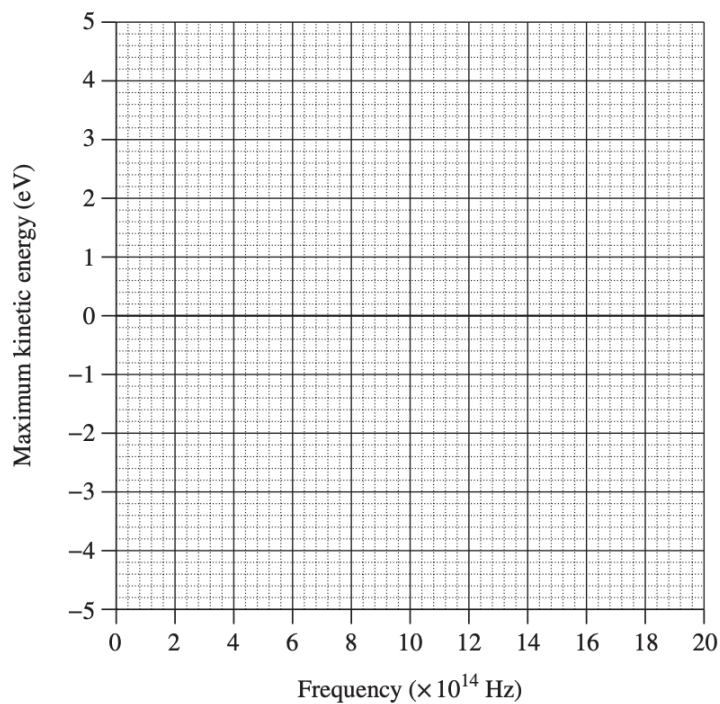
A student investigated the photoelectric effect. The frequency of light incident on a metal surface was varied and the corresponding maximum kinetic energy of the photoelectrons was measured.

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The following results were obtained.

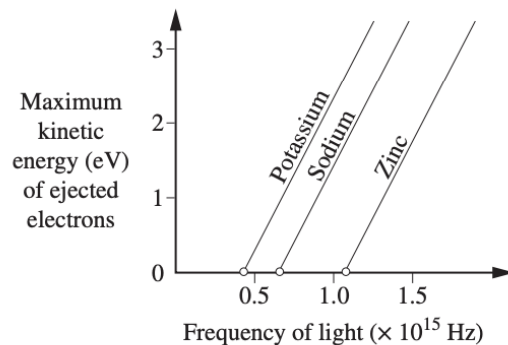
<i>Frequency</i> ($\times 10^{14}$ Hz)	11.2	13.5	15.2	18.6	20.0
<i>Maximum kinetic energy</i> (eV)	0.6	1.3	2.3	3.3	4.2

Plot the results on the axes below and hence determine the work function of the metal in electron volts.



4.4.10 Question 10 (HSC 2018 Q17)

- 17 The graph shows the maximum kinetic energy of electrons ejected from different metals as a function of the frequency of the incident light.



What can be deduced from this graph?

- A. The maximum kinetic energy of ejected electrons is proportional to the number of photons incident on the metal surface.
- B. More photons are required to cause an electron to be ejected from zinc than from potassium.
- C. Any photon that can eject an electron from the surface of zinc must also be able to cause an electron to be ejected from potassium.
- D. For any given frequency that causes electrons to be ejected from all three metals, the number of electrons ejected is always greatest for potassium.

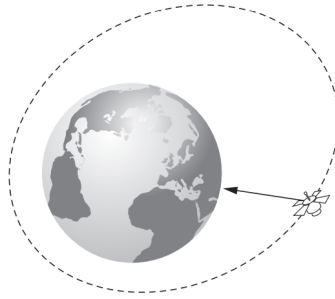
5 BONUS CHALLENGE QUESTIONS

5.0.1 Question 1 (MIT CLASSICAL MECHANICS PS3 Q1)

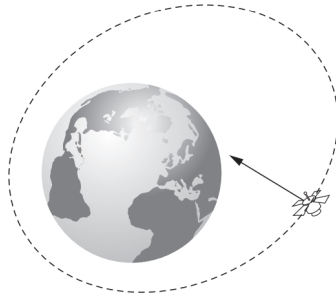
- 1 A satellite orbits Earth as shown.

Which diagram correctly shows the direction of the satellite's acceleration?

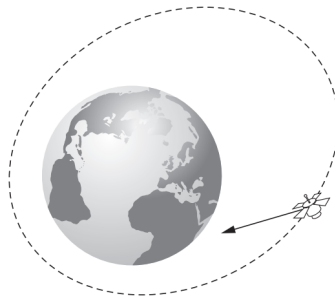
A.



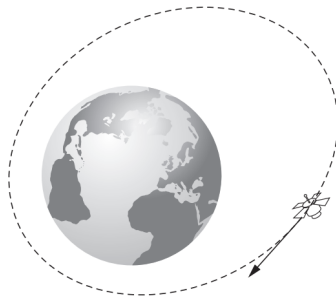
B.



C.

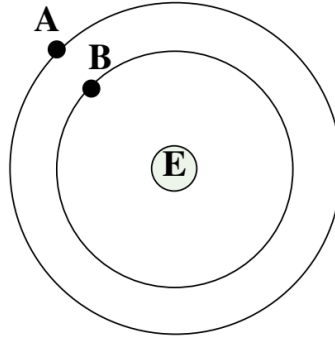


D.



5.0.2 Question 2 (MIT CLASSICAL MECHANICS PS3 Q5)

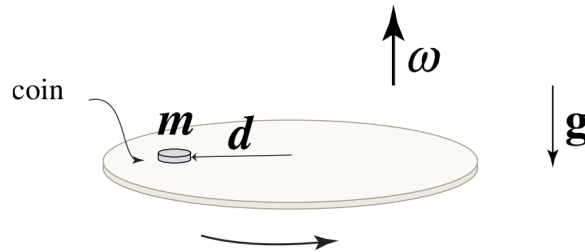
5. Satellite



- (a) Two satellites are orbiting earth at different altitudes. Which satellite orbits at a higher speed v around earth? Assume that the orbits are circular and both satellites have the same mass.
- (b) Which satellite orbits with a longer period, T , around earth? Assume that the orbits are circular and both satellites have the same mass.

5.0.3 Question 3 (MIT CLASSICAL MECHANICS PS3 Q6)

6. A coin on a rotating disk



A coin of mass m is on a rigid disk at a distance d from the center of the disk. There is friction between the coin and the disk. The coefficient of static friction is μ_s . At time $t = 0$, the disk begins to rotate with a constant angular acceleration of magnitude α . The magnitude of the acceleration due to gravity is g .

Express your answers in terms of some or all of the given variables m , d , μ_s , α , t and g as needed.

- (a) While the coin remains at rest relative to the disk, what is f_s , the magnitude of the force of static friction exerted by the disk on the coin as a function of time t ?
- (b) At what angular speed ω will the coin start to slip with respect to the disk?

5.0.4 Question 4 (MIT CLASSICAL MECHANICS PS7 Q1)

Problem Set 7

1. Object Sliding Down an Inclined Plane

An object of mass $m = 4.0$ kg, starting from rest, slides down an inclined plane of length $l = 3.0$ m. The plane is inclined by an angle of $\theta = 30^\circ$ to the ground. The coefficient of kinetic friction $\mu_k = 0.2$. At the bottom of the plane, the mass slides along a rough surface with a coefficient of kinetic friction $\mu_k = 0.3$ until it comes to rest. The goal of this problem is to find out how far the object slides along the rough surface.



- What is the work done by the friction force while the mass is sliding down the inclined plane? (Is it positive or negative?)
- What is the work done by the gravitational force while the mass is sliding down the inclined plane? (Is it positive or negative?)
- What is the kinetic energy of the mass when it just reaches the bottom of the inclined plane?
- Symbolically, what is the work done by the friction force while the mass is sliding along the ground? Is this positive or negative? Express your answer in terms of some or all of the following: m , μ_k , g , and d where d is the distance it takes the object to stop measured from the bottom of the incline.
- How far from the bottom of the inclined plane does the object slide along the rough surface?